Single Column Model (SCM) Diagnostic Runs

GSPT Simulation Pipeline

July 15, 2026

1 Overview

This document compiles every current SCM case report under results/ that has a compiled wrapper PDF.

2 Case 1: gabls1 80x2m 0h report

3 SCM Case Summary: gabls1

This section was generated automatically from the SCM diagnostics artifact `results/gabls1/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PREC Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.` and returned `Success` .

3.1 Core Run Metadata

- Case identifier: `gabls1`
- Output directory: `results/gabls1`
- Number of saved time samples: 13
- Number of profile snapshots: 1
- Solver accepted/rejected steps: 62 / 1
- RHS evaluations: 15572
- Solver tolerances: abstol $= 1.000 \times 10^{-8}$, reltol $= 1.000 \times 10^{-6}$

3.2 Forcing and Boundary Conditions

Table 1: Forcing and Boundary Condition Parameters

Parameter	**Symbol**	**Value**
Zonal geostrophic wind	U_g	$8.000000\,\mathrm{m\,s^{-1}}$
Meridional geostrophic wind	V_g	$0.000000\,\mathrm{m\,s^{-1}}$
Surface roughness (momentum)	z_{0m}	n/a m
Surface roughness (heat)	z_{0h}	n/a m
Coriolis parameter	f	1.000e-04 $\mathrm{s^{-1}}$
Reference potential temperature	θ_a	265.000000 K
Deep soil temperature	T_{deep}	262.000000 K
Downward longwave forcing	$R_{\downarrow}$	$245.000000\,\mathrm{W\,m^{-2}}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{\mathrm{min,surf}}$	$0.001000\,\mathrm{m^2\,s^{-1}}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	$0.000000\,\mathrm{s^{-1}}$

3.3 Numerical Verification

- Maximum surface energy closure error: $0\,\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.103721, 79.407743]\ \mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [0.119669, 36.559392]$

3.4 Generated Artifacts

- Time-series CSV: `results/gabls1/time_series.csv`
- Diagnostic summary JSON: `results/gabls1/summary.json`
- Payload JLD2: `results/gabls1/payload.jld2`

3.5 Figure Manifest

1. Time series of T_s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times
5. Surface energy budget components (R_n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e_xi)
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

3.6 Included Plots

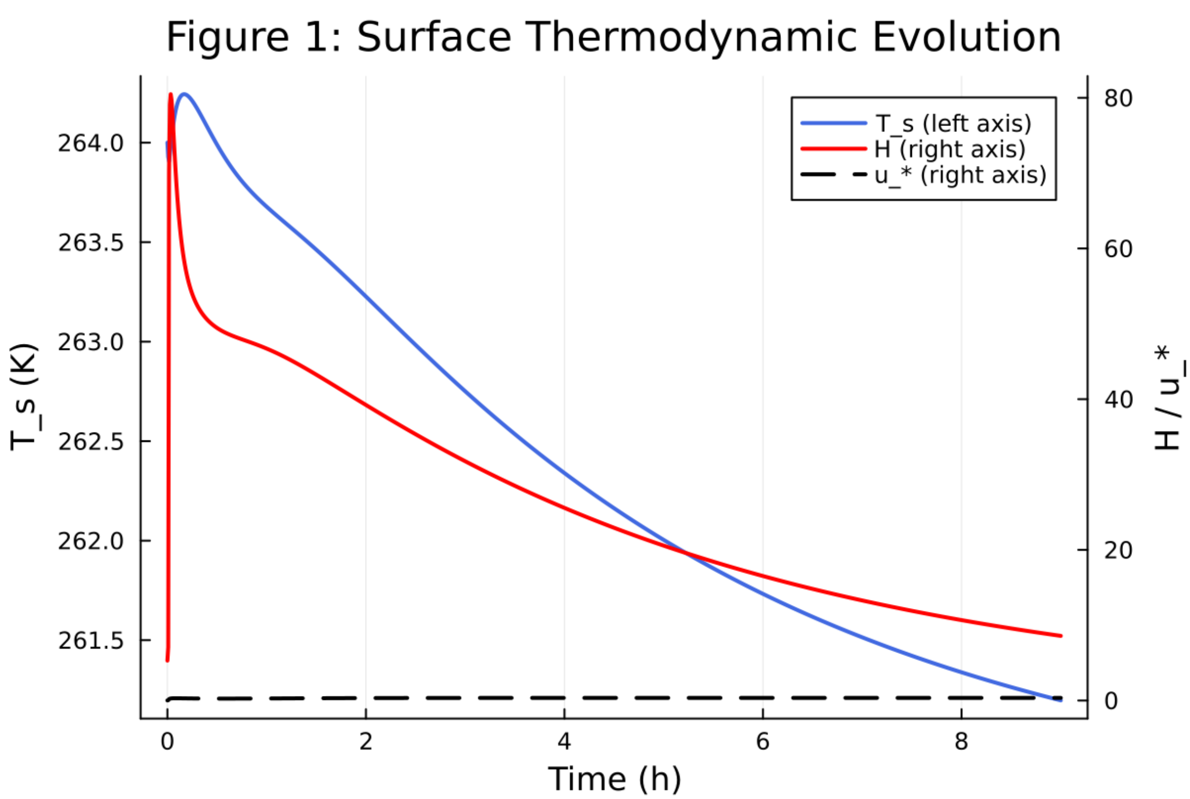

Figure 1: Time series of T_s, sensible heat flux, and friction velocity

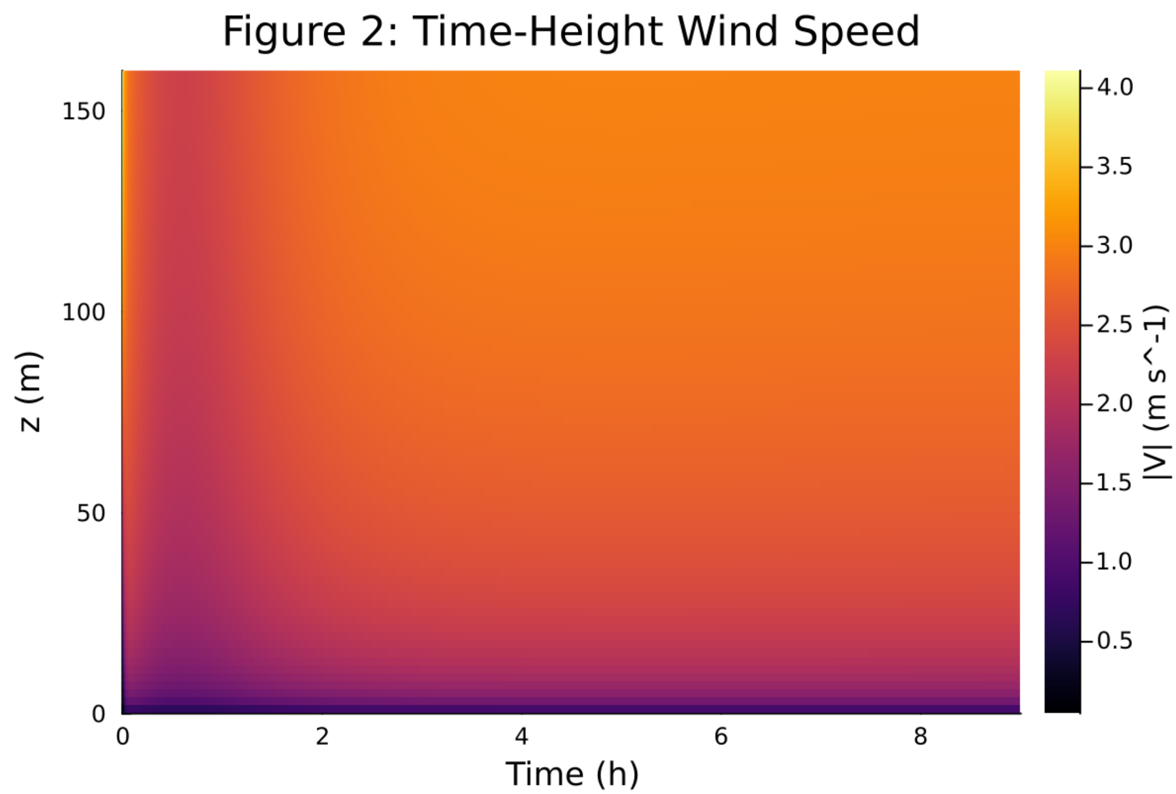

Figure 2: Time-height contour of wind speed with LLJ evolution

4 Case 2: gabls1 80x2m 9h report

5 SCM Case Summary: gabls1

This section was generated automatically from the SCM diagnostics artifact `results/gabls1/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PREC Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.t` and returned `Success` .

5.1 Core Run Metadata

- Case identifier: `gabls1`
- Output directory: `results/gabls1`
- Number of saved time samples: 1081
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 416 / 1
- RHS evaluations: 104426
- Solver tolerances: abstol $= 1.000 \times 10^{-8}$, reltol $= 1.000 \times 10^{-6}$

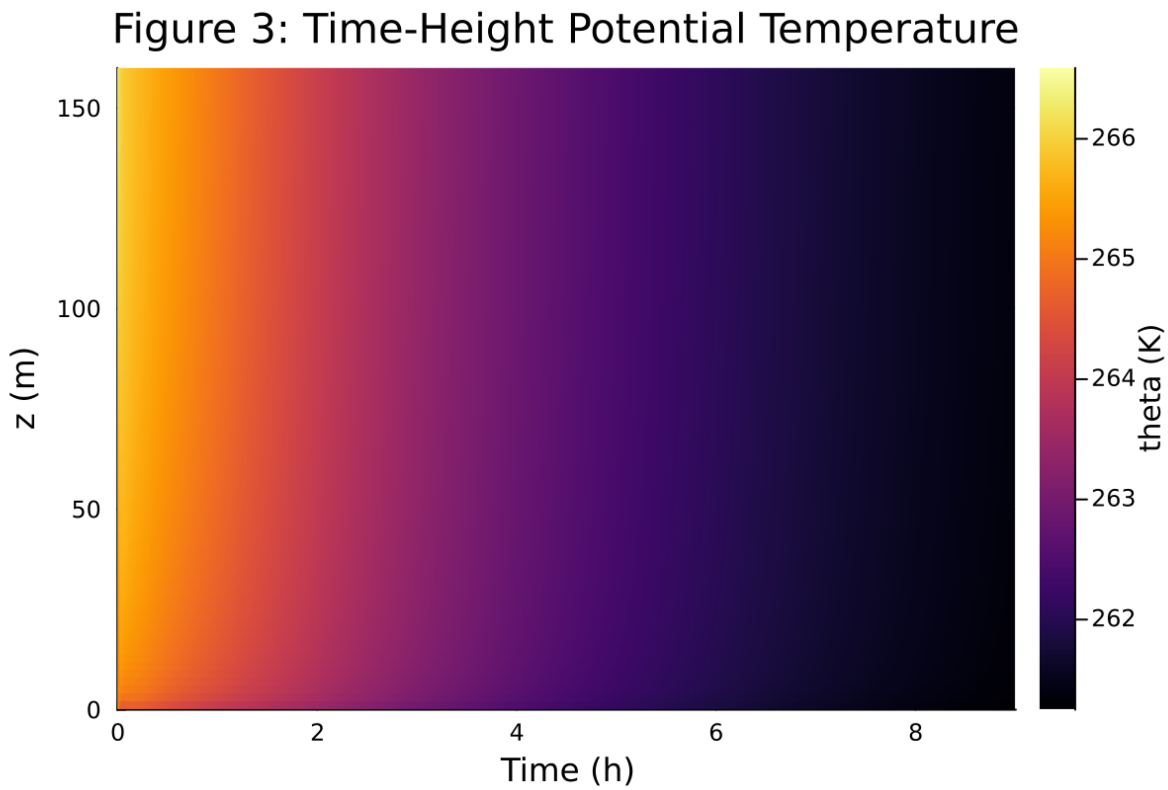

Figure 3: Time-height contour of potential temperature and inversion growth

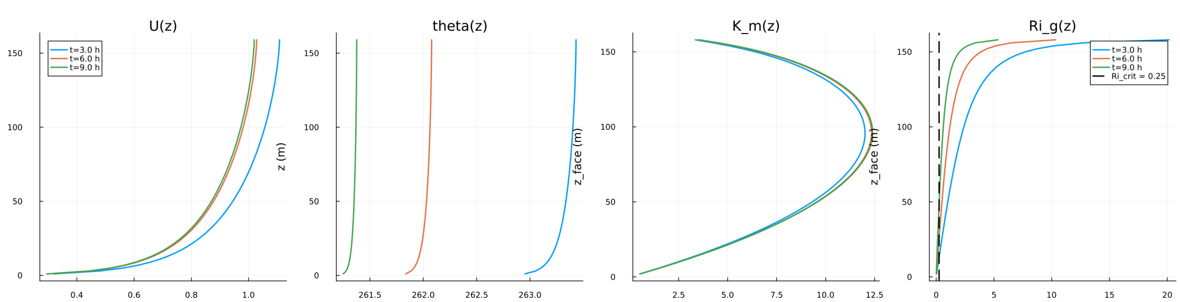

Figure 4: Vertical profiles of U, theta, and Km at representative times

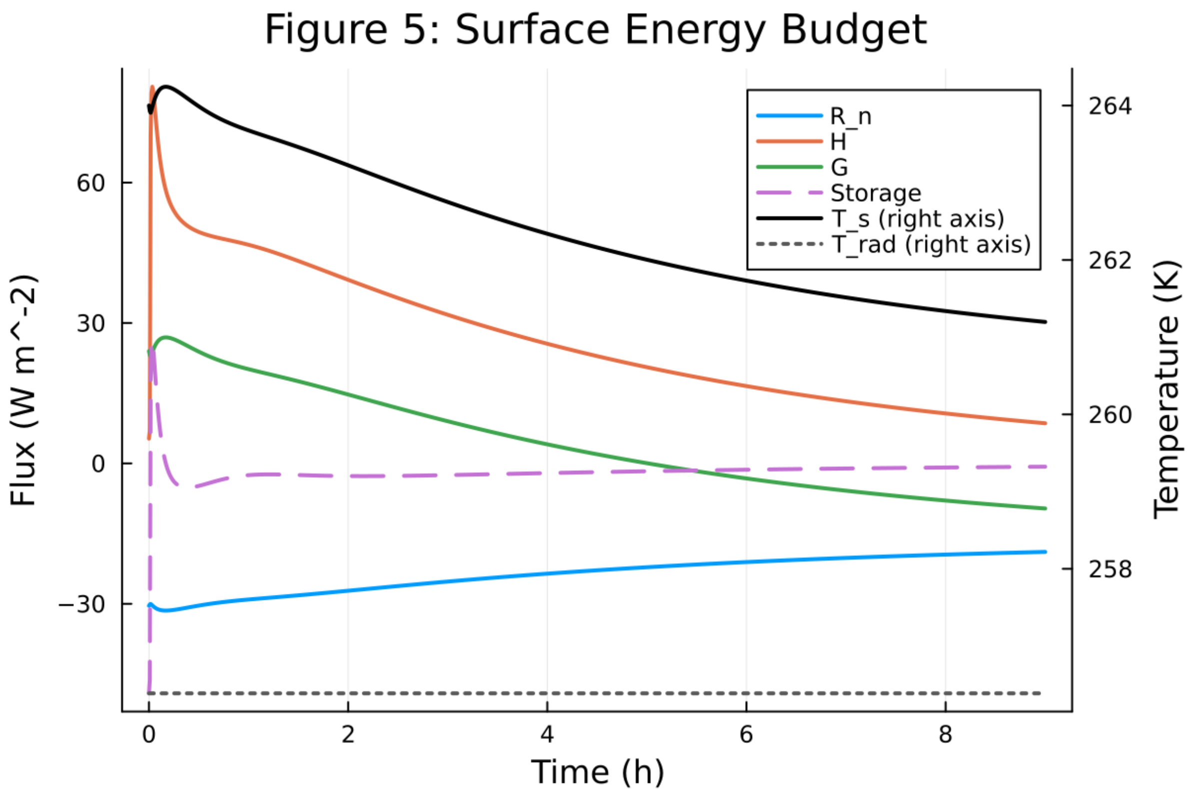

Figure 5: Surface energy budget components (R_n, H, G, storage)

Table 2: Forcing and Boundary Condition Parameters

Parameter	**Symbol**	**Value**
Zonal geostrophic wind	U_g	$8.000000\,\mathrm{m\,s^{-1}}$
Meridional geostrophic wind	V_g	$0.000000\,\mathrm{m\,s^{-1}}$
Surface roughness (momentum)	z_{0m}	n/a m
Surface roughness (heat)	z_{0h}	n/a m
Coriolis parameter	f	1.000e-04 $\mathrm{s^{-1}}$
Reference potential temperature	θ_a	265.000000 K
Deep soil temperature	T_{deep}	262.000000 K
Downward longwave forcing	$R_{\downarrow}$	$245.000000\,\mathrm{W\,m^{-2}}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{\mathrm{min,surf}}$	$0.001000\,\mathrm{m^2\,s^{-1}}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	$0.000000\,\mathrm{s^{-1}}$

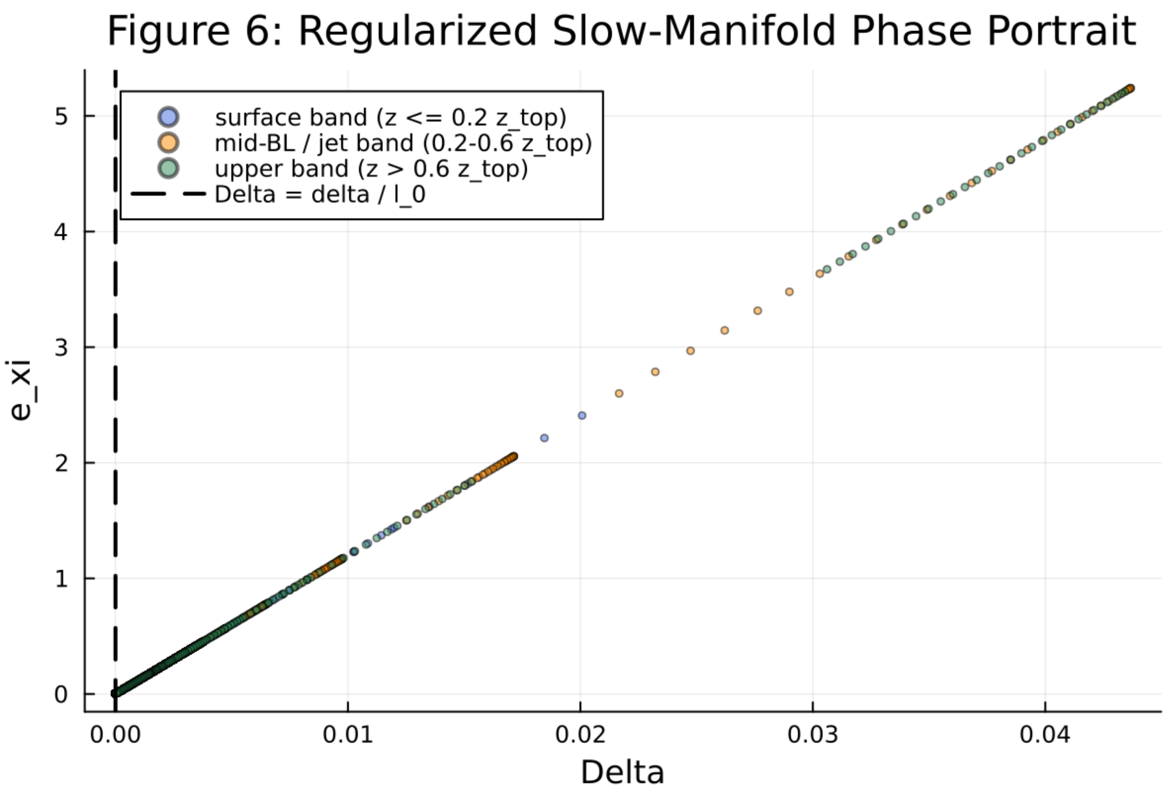

Figure 6: Phase portrait on regularized manifold (Delta vs e_xi)

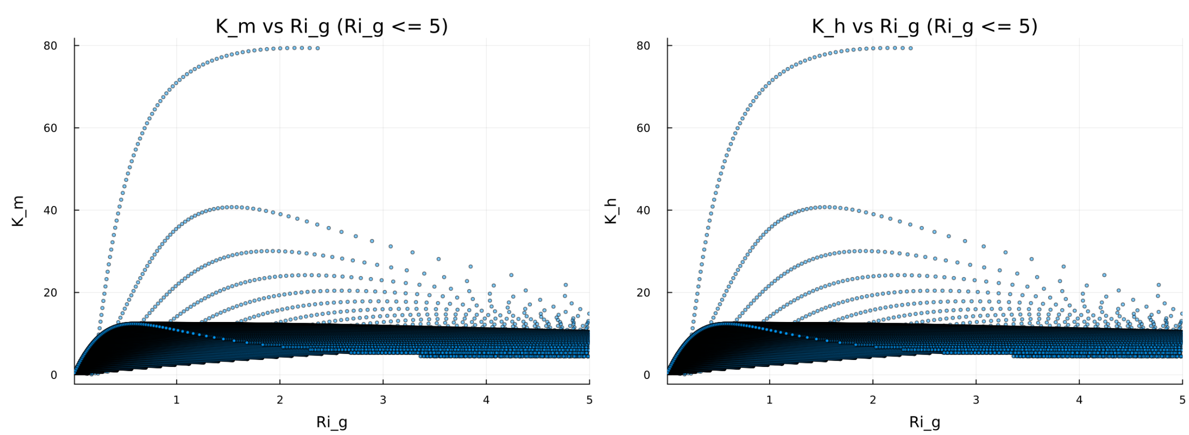

Figure 7: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

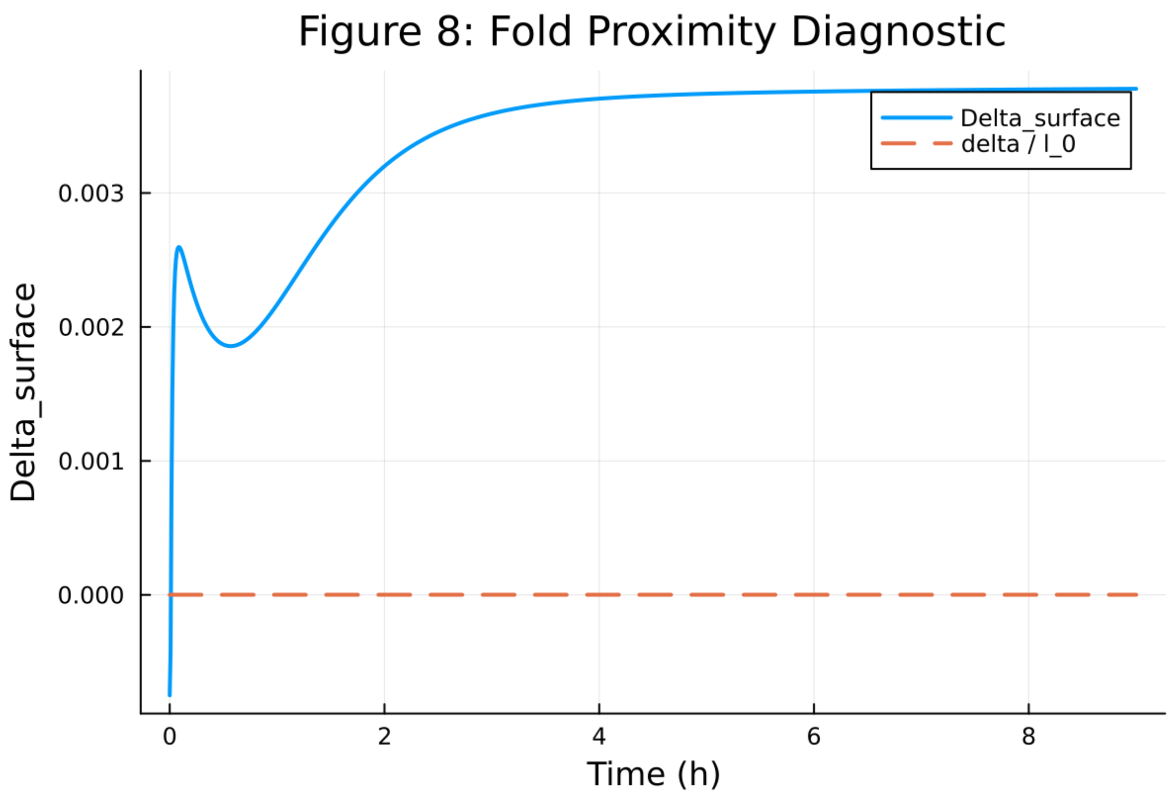

Figure 8: Fold proximity diagnostics versus time

5.2 Forcing and Boundary Conditions

5.3 Numerical Verification

- Maximum surface energy closure error: $0\,\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.103721, 79.407743]\ \mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [0.009840, 82.319866]$

5.4 Generated Artifacts

- Time-series CSV: `results/gabls1/time_series.csv`
- Diagnostic summary JSON: `results/gabls1/summary.json`
- Payload JLD2: `results/gabls1/payload.jld2`

5.5 Figure Manifest

1. Time series of T_s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times

5. Surface energy budget components (R__n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e__xi)
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

5.6 Included Plots

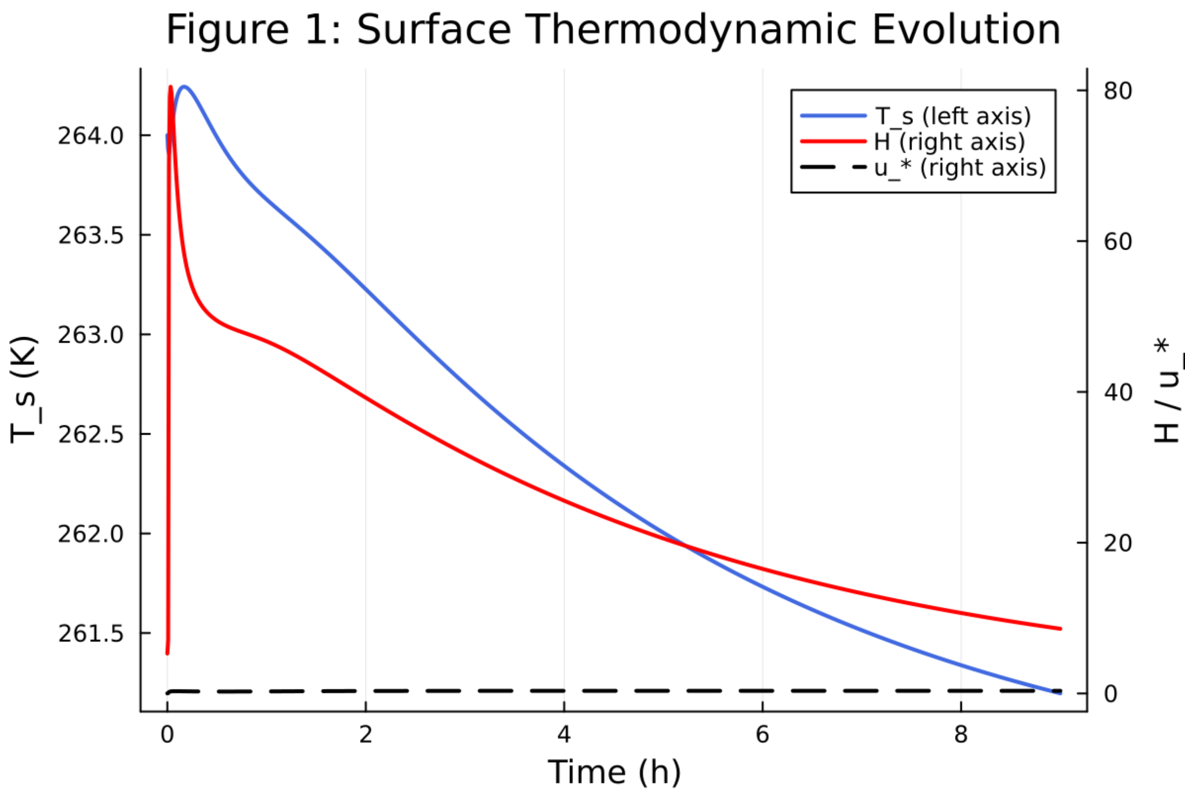

Figure 9: Time series of T__s, sensible heat flux, and friction velocity

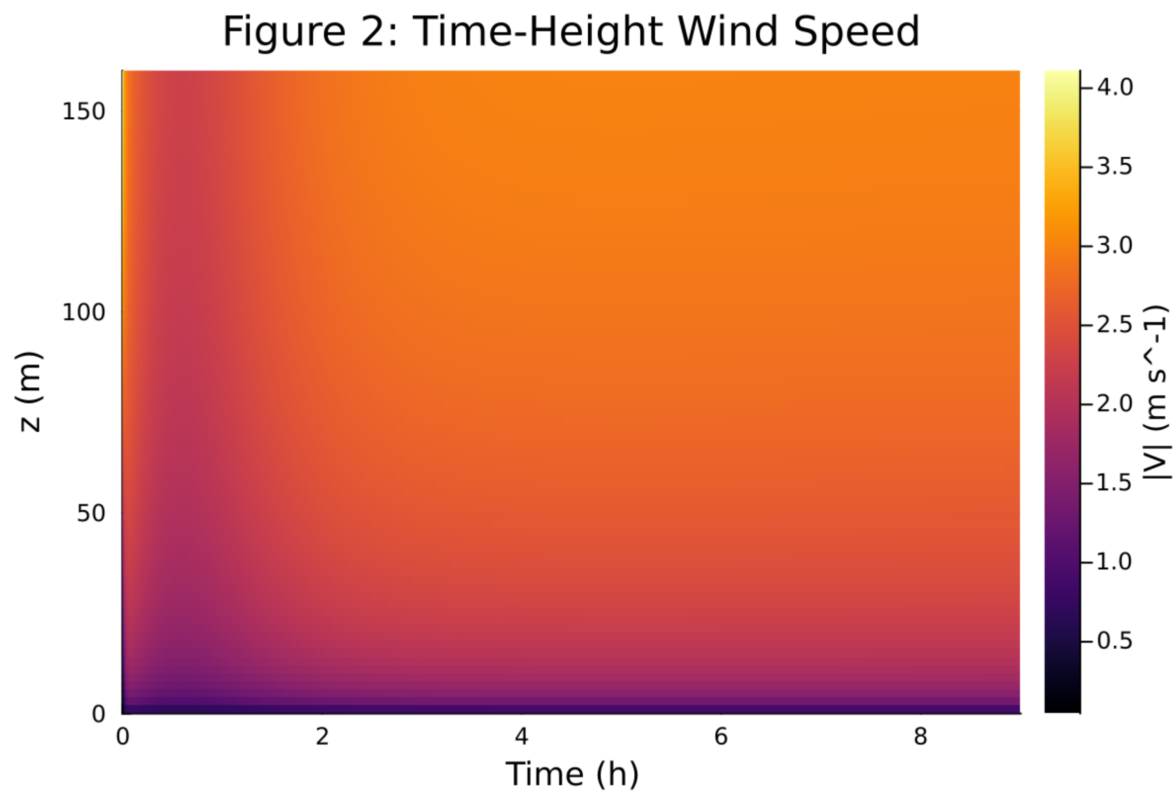

Figure 10: Time-height contour of wind speed with LLJ evolution

6 Case 3: scm case report

7 SCM Case Summary: gabls1

This section was generated automatically from the SCM diagnostics artifact `results/gabls1/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PREC Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.t` and returned `Success` .

7.1 Core Run Metadata

- Case identifier: `gabls1`
- Output directory: `results/gabls1`
- Number of saved time samples: 1081
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 416 / 1
- RHS evaluations: 104426
- Solver tolerances: abstol $= 1.000 \times 10^{-8}$, reltol $= 1.000 \times 10^{-6}$

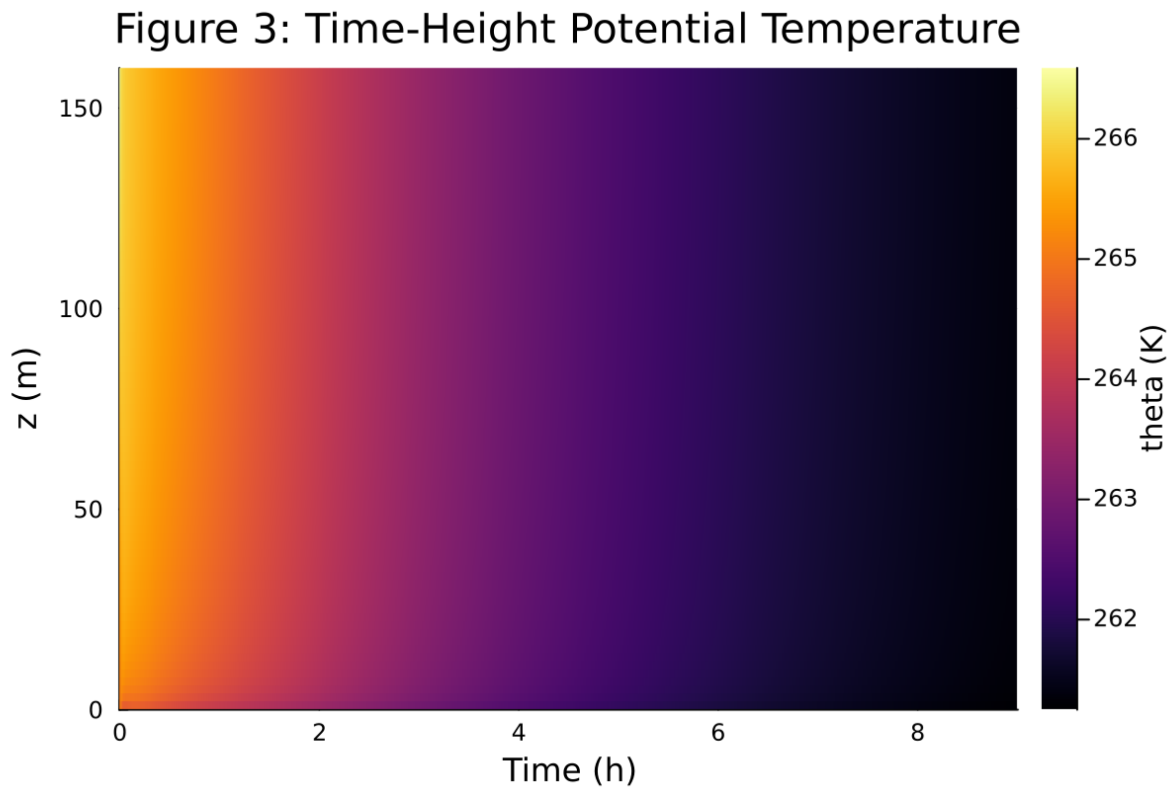

Figure 11: Time-height contour of potential temperature and inversion growth

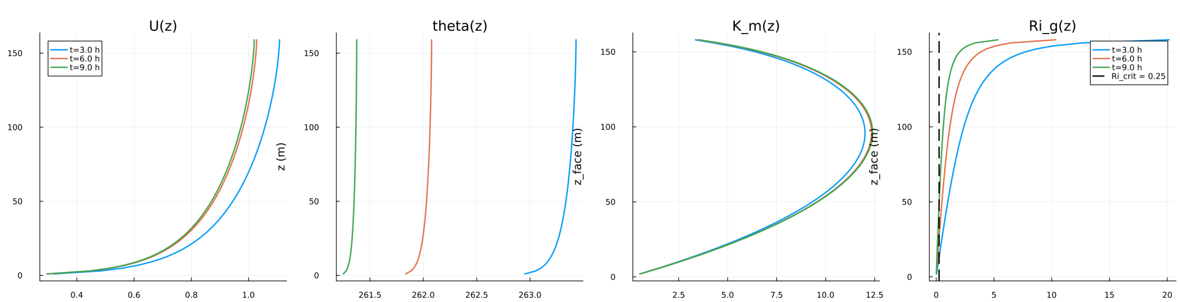

Figure 12: Vertical profiles of U, theta, and Km at representative times

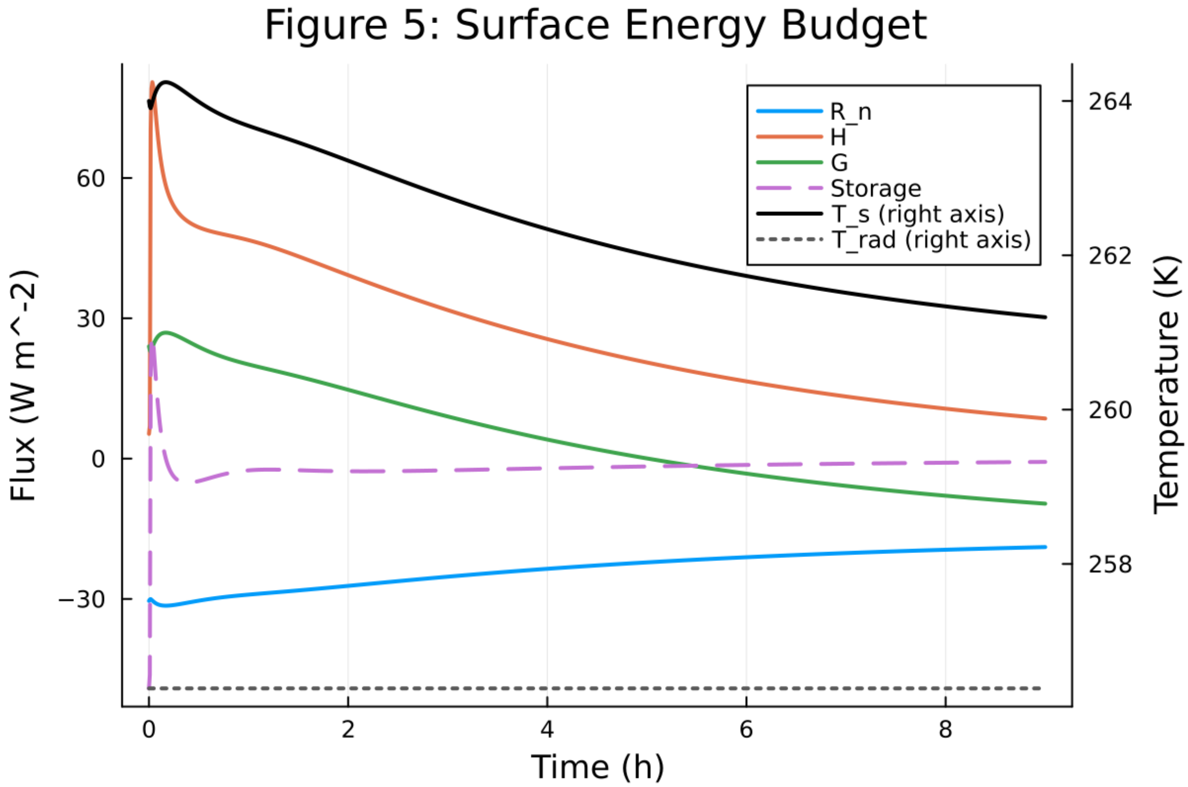

Figure 13: Surface energy budget components (R_n, H, G, storage)

Table 3: Forcing and Boundary Condition Parameters

Parameter	**Symbol**	**Value**
Zonal geostrophic wind	U_g	$8.000000\,\mathrm{m\,s^{-1}}$
Meridional geostrophic wind	V_g	$0.000000\,\mathrm{m\,s^{-1}}$
Surface roughness (momentum)	z_{0m}	n/a m
Surface roughness (heat)	z_{0h}	n/a m
Coriolis parameter	f	$1.000\text{e-}04\,\mathrm{s^{-1}}$
Reference potential temperature	θ_a	$265.000000\,\mathrm{K}$
Deep soil temperature	T_{deep}	$262.000000\,\mathrm{K}$
Downward longwave forcing	$R_\downarrow$	$245.000000\,\mathrm{W\,m^{-2}}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	$0.100000\,\mathrm{m}$
Surface diffusivity floor	$k_{\mathrm{min,surf}}$	$0.001000\,\mathrm{m^2\,s^{-1}}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	$265.000000\,\mathrm{K}$
Top relaxation coefficient	λ_{top}	$0.000000\,\mathrm{s^{-1}}$

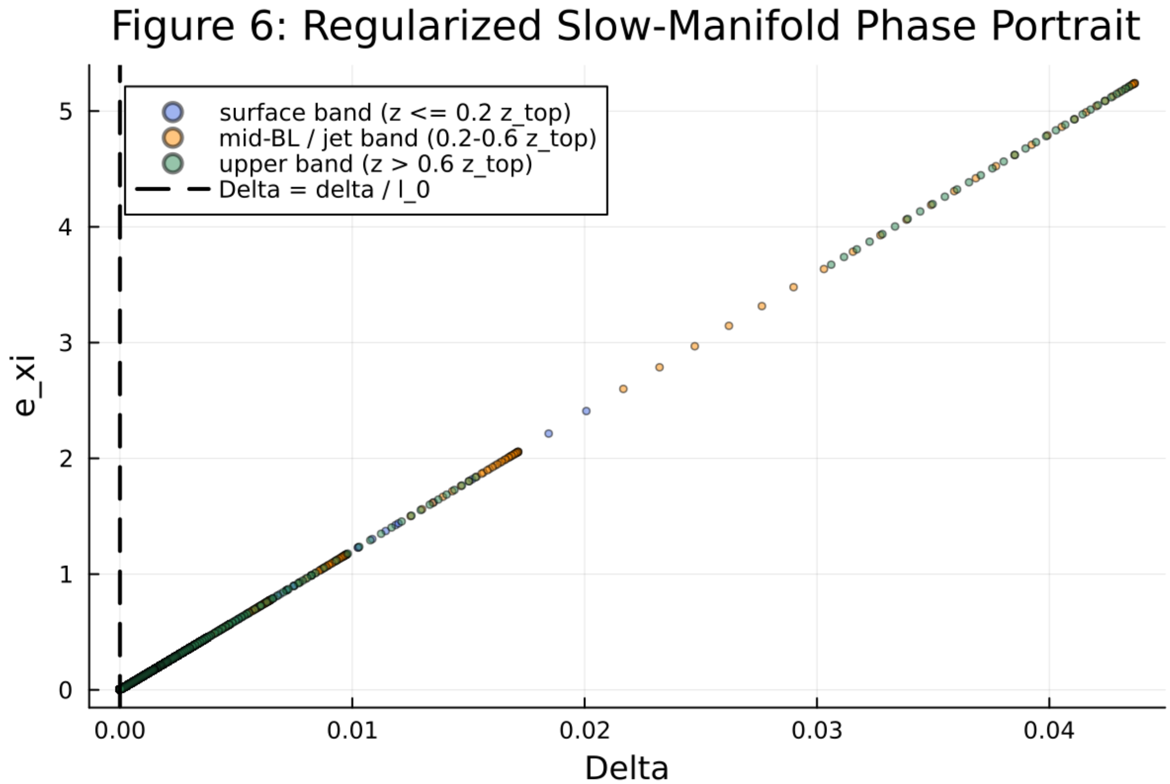

Figure 14: Phase portrait on regularized manifold (Delta vs e__xi)

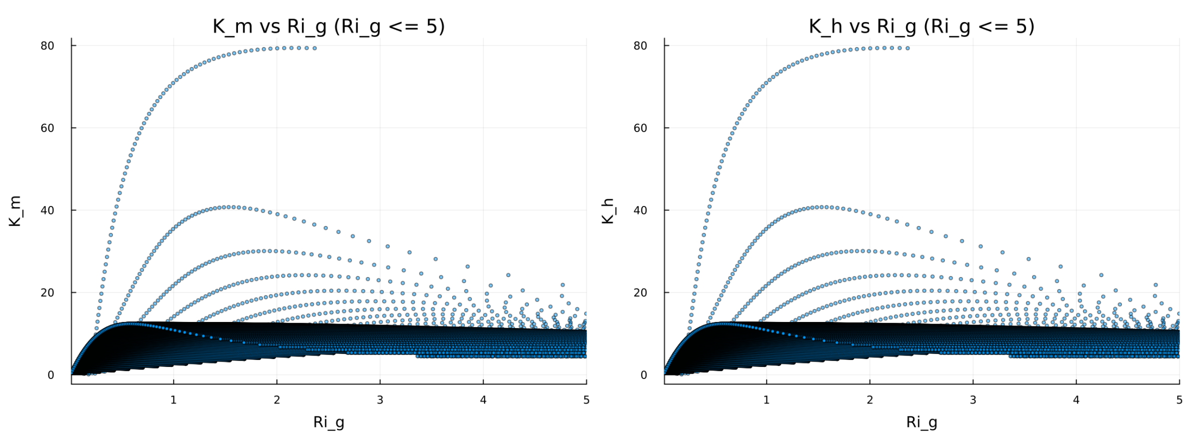

Figure 15: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

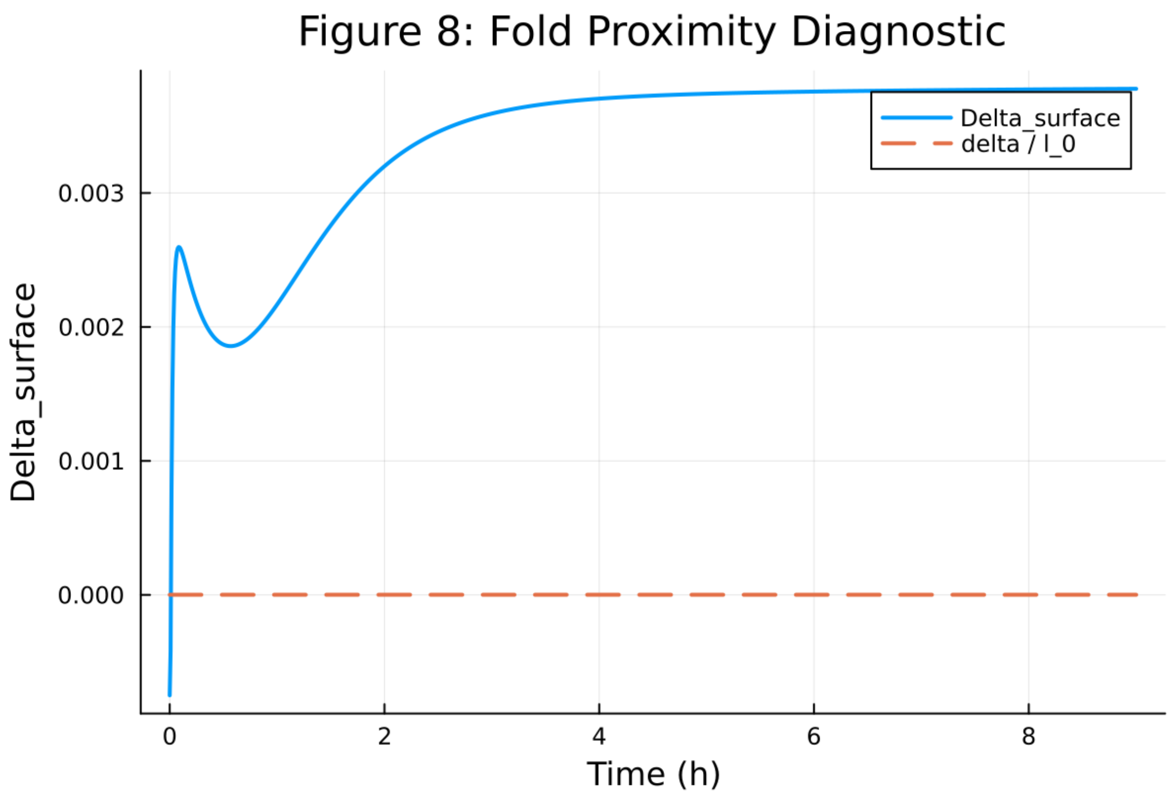

Figure 16: Fold proximity diagnostics versus time

7.2 Forcing and Boundary Conditions

7.3 Numerical Verification

- Maximum surface energy closure error: $0\,\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.103721, 79.407743]\ \mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [0.009840, 82.319866]$

7.4 Generated Artifacts

- Time-series CSV: `results/gabls1/time_series.csv`
- Diagnostic summary JSON: `results/gabls1/summary.json`
- Payload JLD2: `results/gabls1/payload.jld2`

7.5 Figure Manifest

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8. Fold proximity diagnostics versus time

7.6 Included Plots

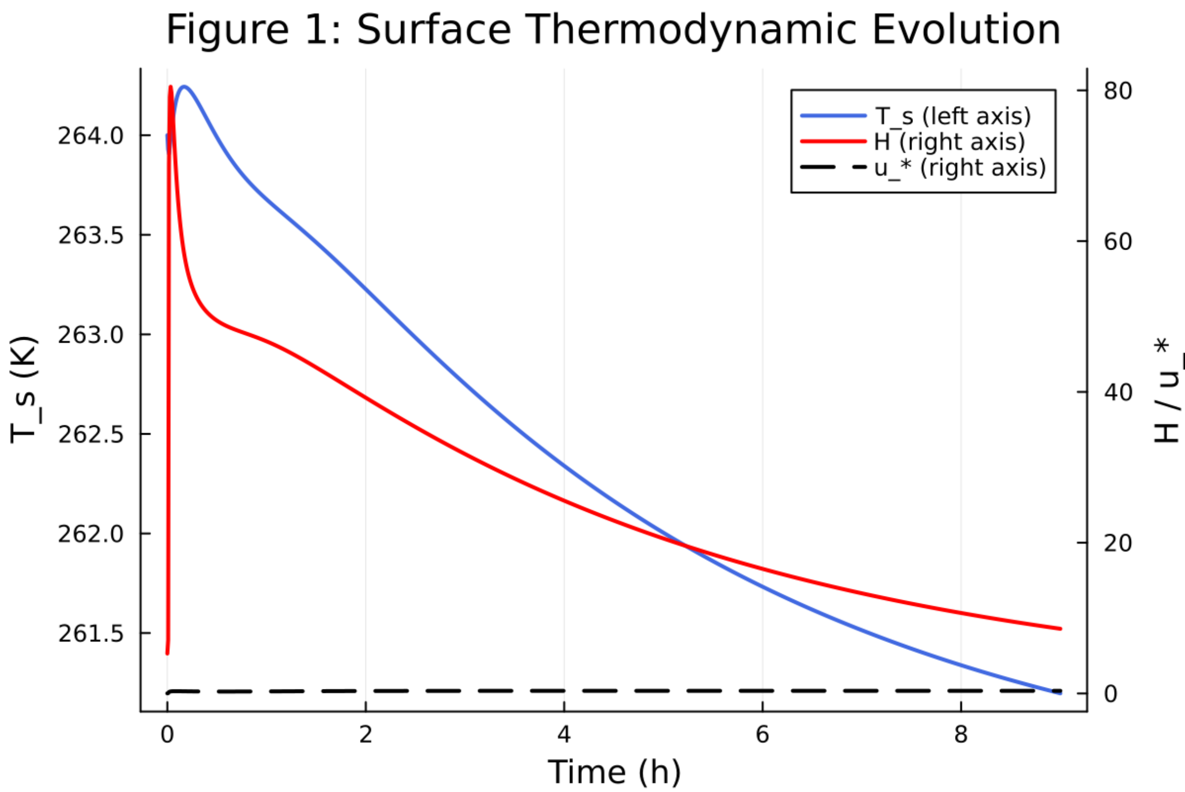

Figure 17: Time series of T_s, sensible heat flux, and friction velocity

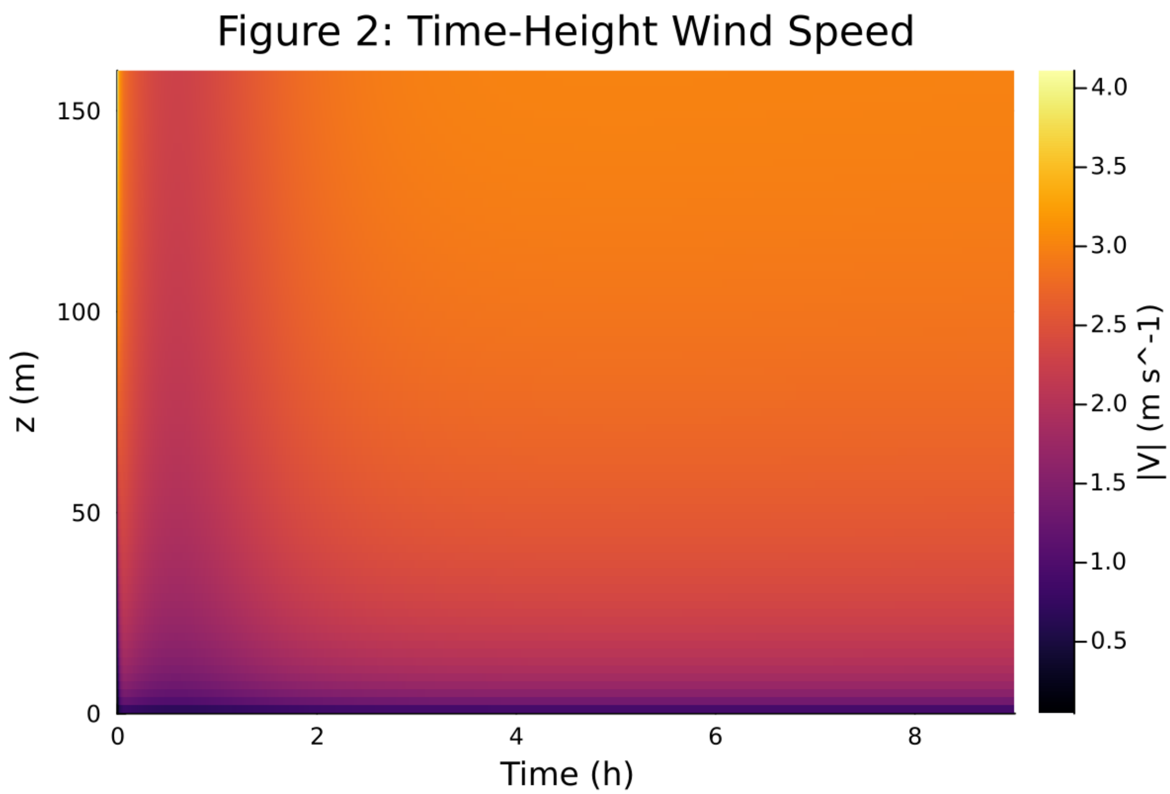

Figure 18: Time-height contour of wind speed with LLJ evolution

8 Case 4: idealized sbl 80x2m 9h report

9 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/idealized_sbl/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PREC Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.t` and returned `Success` .

9.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/idealized_sbl`
- Number of saved time samples: 1081
- Number of profile snapshots: 4
- Solver accepted/rejected steps: 438 / 2
- RHS evaluations: 109956
- Solver tolerances: abstol = 1.000×10^{-8}, reltol = 1.000×10^{-6}

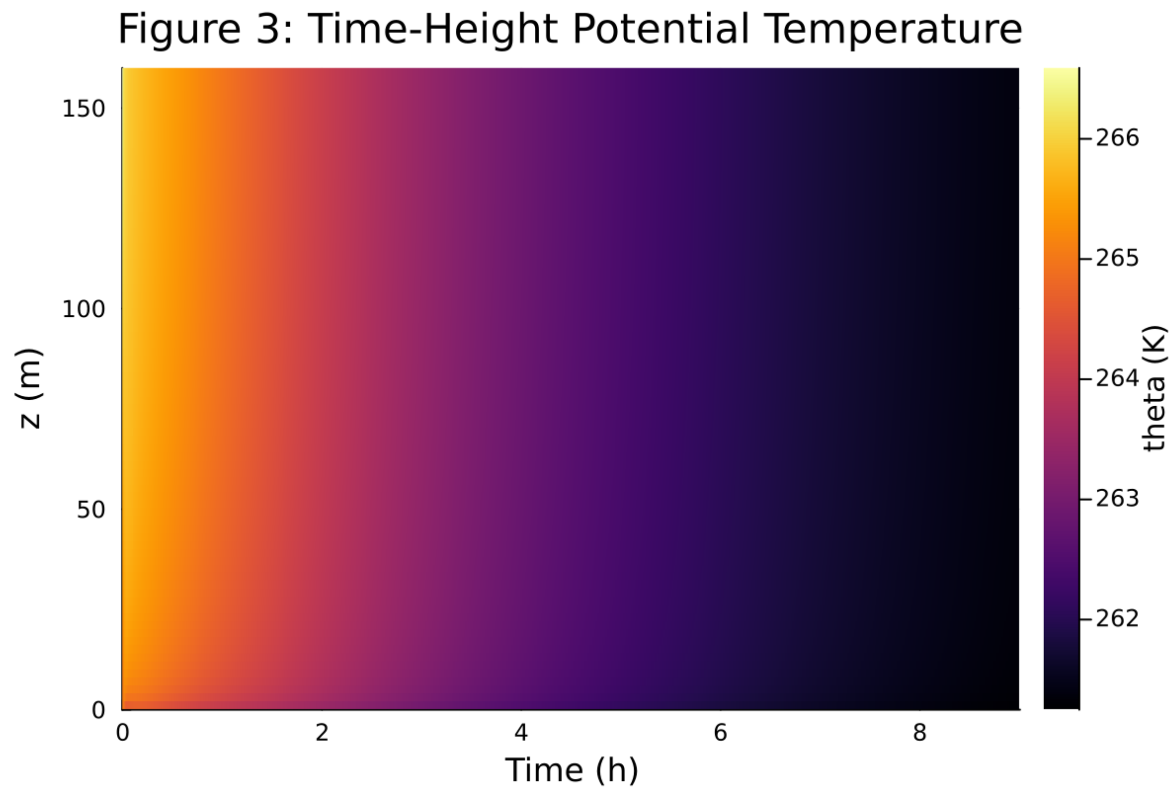

Figure 19: Time-height contour of potential temperature and inversion growth

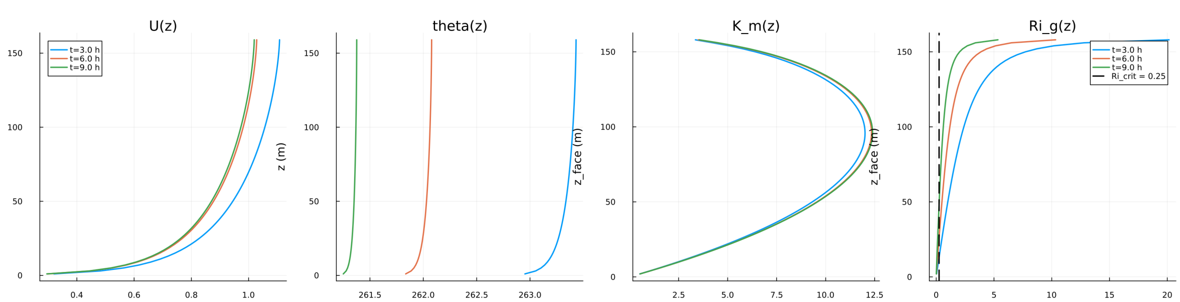

Figure 20: Vertical profiles of U, theta, and Km at representative times

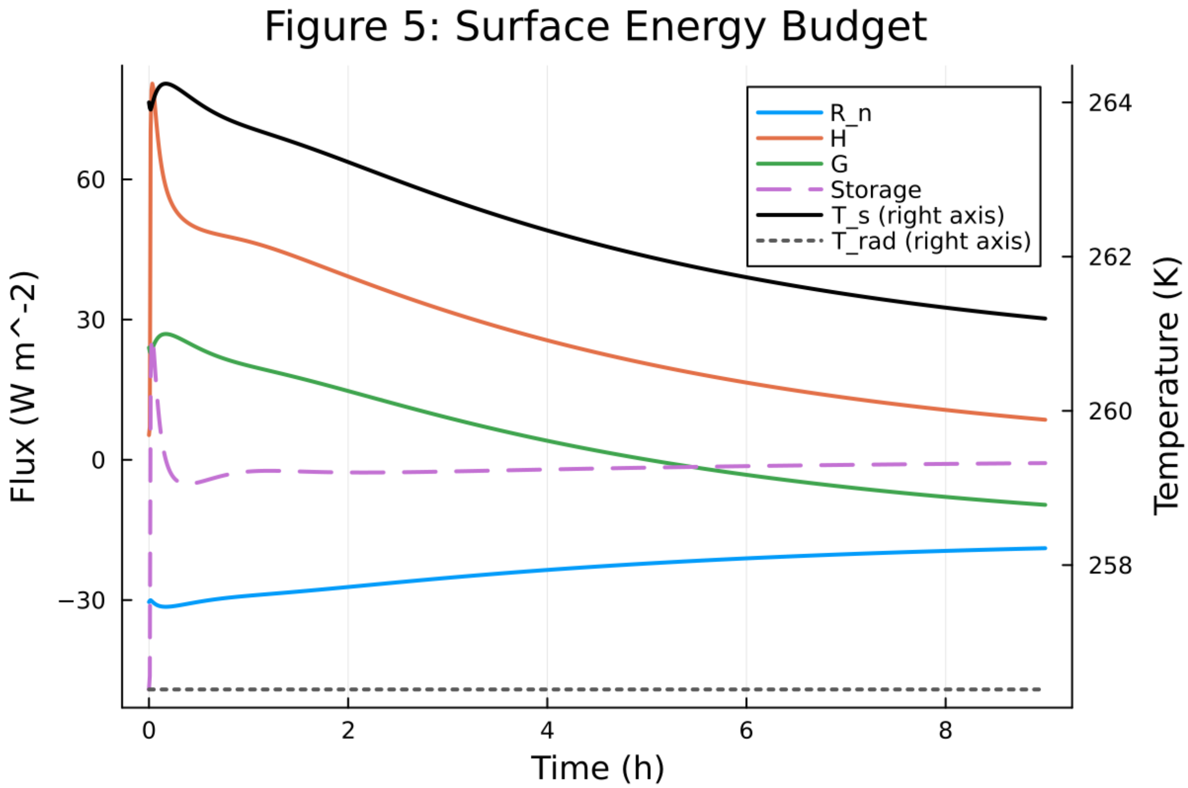

Figure 21: Surface energy budget components (R_n, H, G, storage)

Table 4: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	$10.000000\,\mathrm{m\,s^{-1}}$
Meridional geostrophic wind	V_g	$0.000000\,\mathrm{m\,s^{-1}}$
Surface roughness (momentum)	z_{0m}	n/a m
Surface roughness (heat)	z_{0h}	n/a m
Coriolis parameter	f	1.000e-04 $\mathrm{s^{-1}}$
Reference potential temperature	θ_a	280.000000 K
Deep soil temperature	T_{deep}	276.000000 K
Downward longwave forcing	$R_{\downarrow}$	$300.000000\,\mathrm{W\,m^{-2}}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{\mathrm{min,surf}}$	$0.001000\,\mathrm{m^2\,s^{-1}}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	$0.000000\,\mathrm{s^{-1}}$

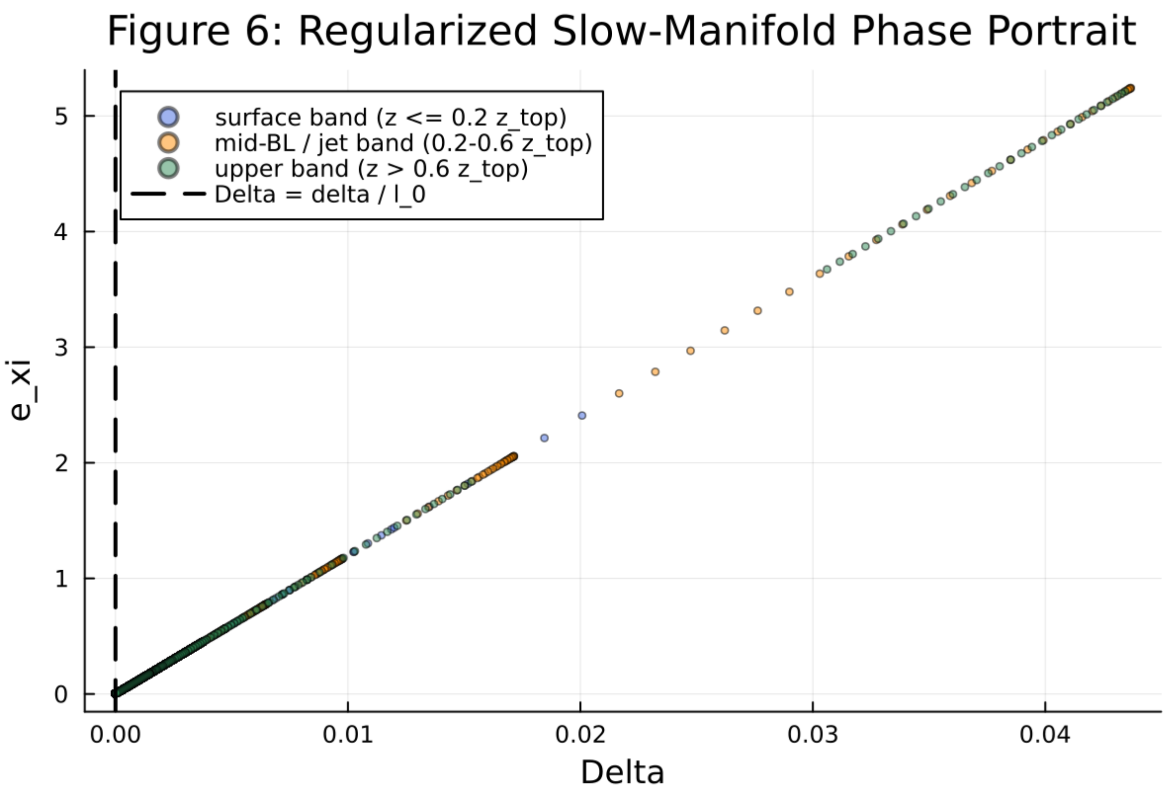

Figure 22: Phase portrait on regularized manifold (Delta vs e__xi)

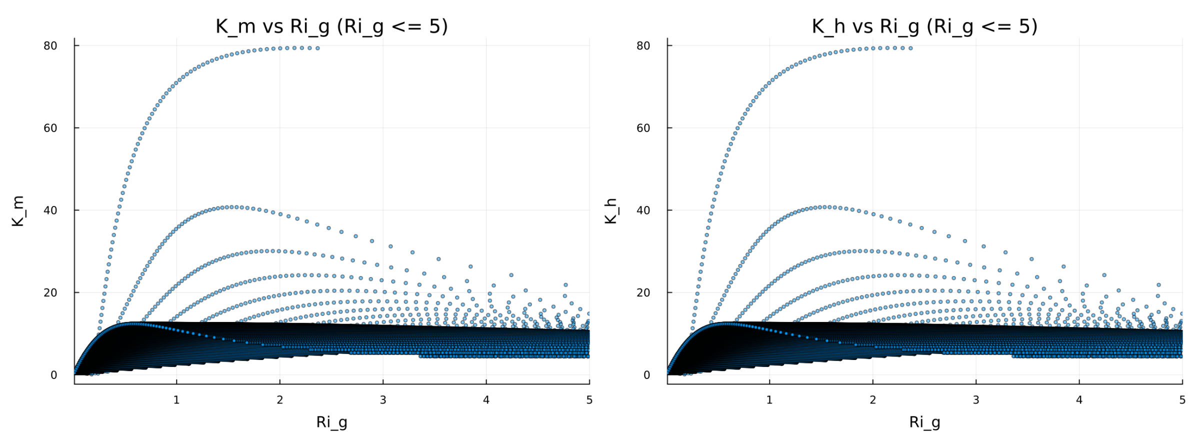

Figure 23: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

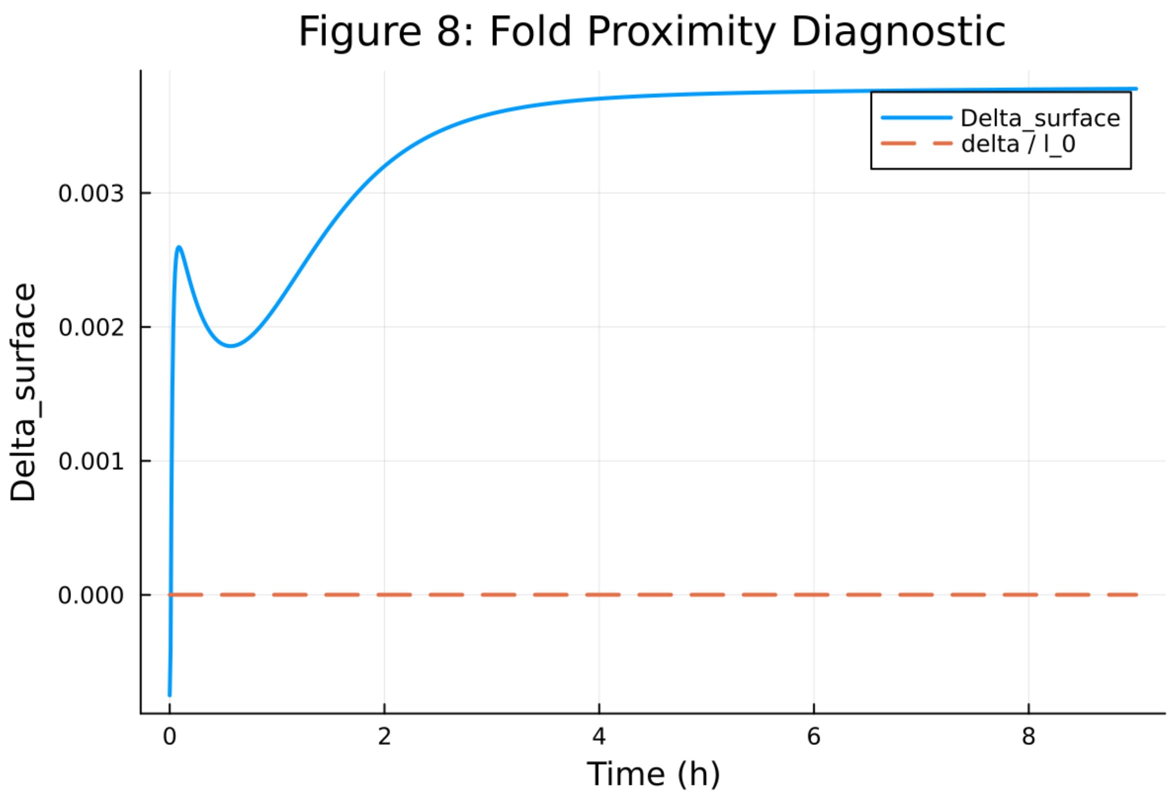

Figure 24: Fold proximity diagnostics versus time

9.2 Forcing and Boundary Conditions

9.3 Numerical Verification

- Maximum surface energy closure error: $0\,\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.184562, 108.426452]\ \mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [0.007283, 43.999865]$

9.4 Generated Artifacts

- Time-series CSV: `results/idealized_sbl/time_series.csv`
- Diagnostic summary JSON: `results/idealized_sbl/summary.json`
- Payload JLD2: `results/idealized_sbl/payload.jld2`

9.5 Figure Manifest

1. Time series of T_s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times

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6. Phase portrait on regularized manifold (Delta vs e__xi)
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

9.6 Included Plots

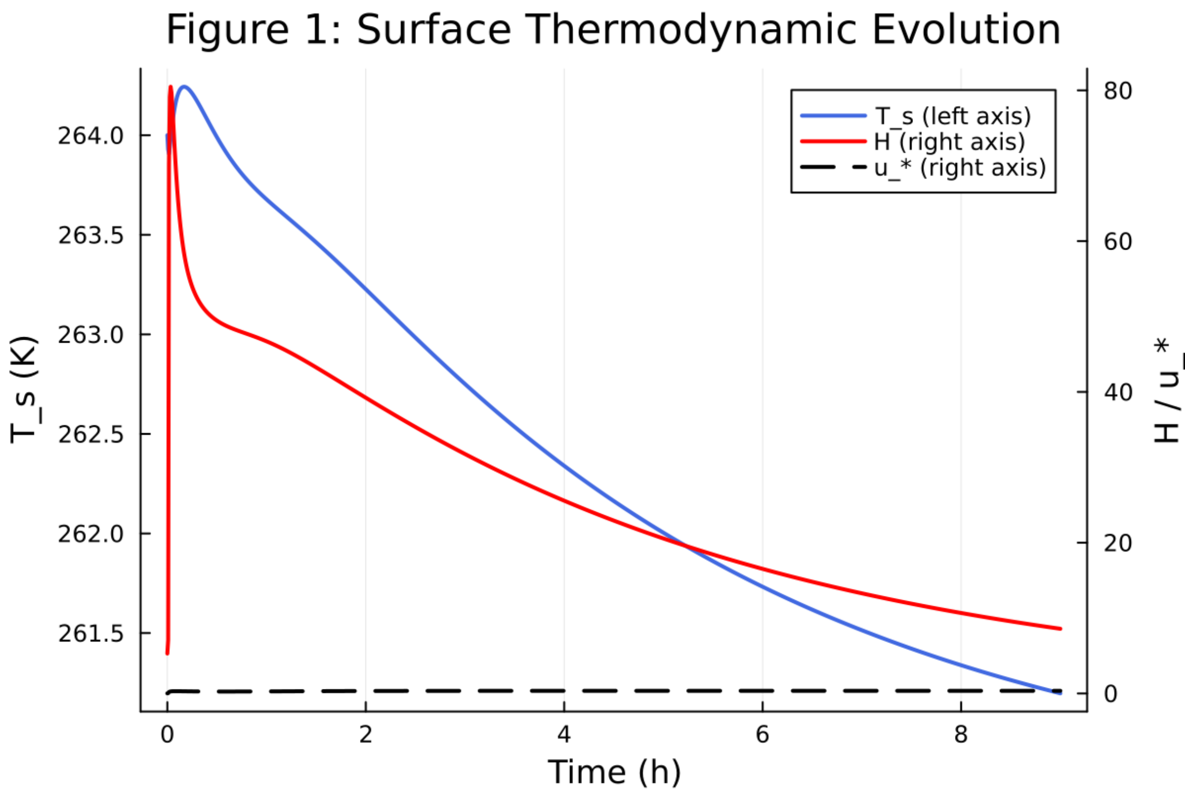

Figure 25: Time series of T__s, sensible heat flux, and friction velocity

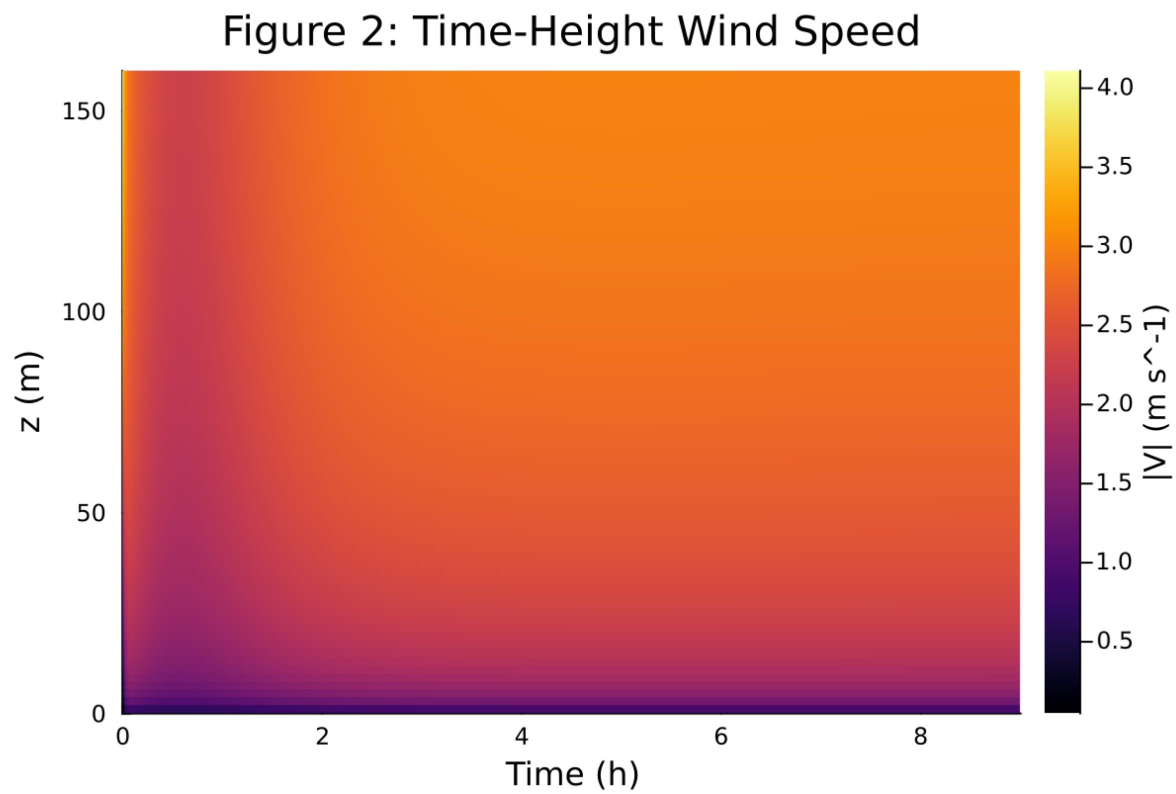

Figure 26: Time-height contour of wind speed with LLJ evolution

10 Case 5: scm case report

11 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/idealized_sbl/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PREC Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.t` and returned `Success` .

11.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/idealized_sbl`
- Number of saved time samples: 1081
- Number of profile snapshots: 4
- Solver accepted/rejected steps: 438 / 2
- RHS evaluations: 109956
- Solver tolerances: abstol $= 1.000 \times 10^{-8}$, reltol $= 1.000 \times 10^{-6}$

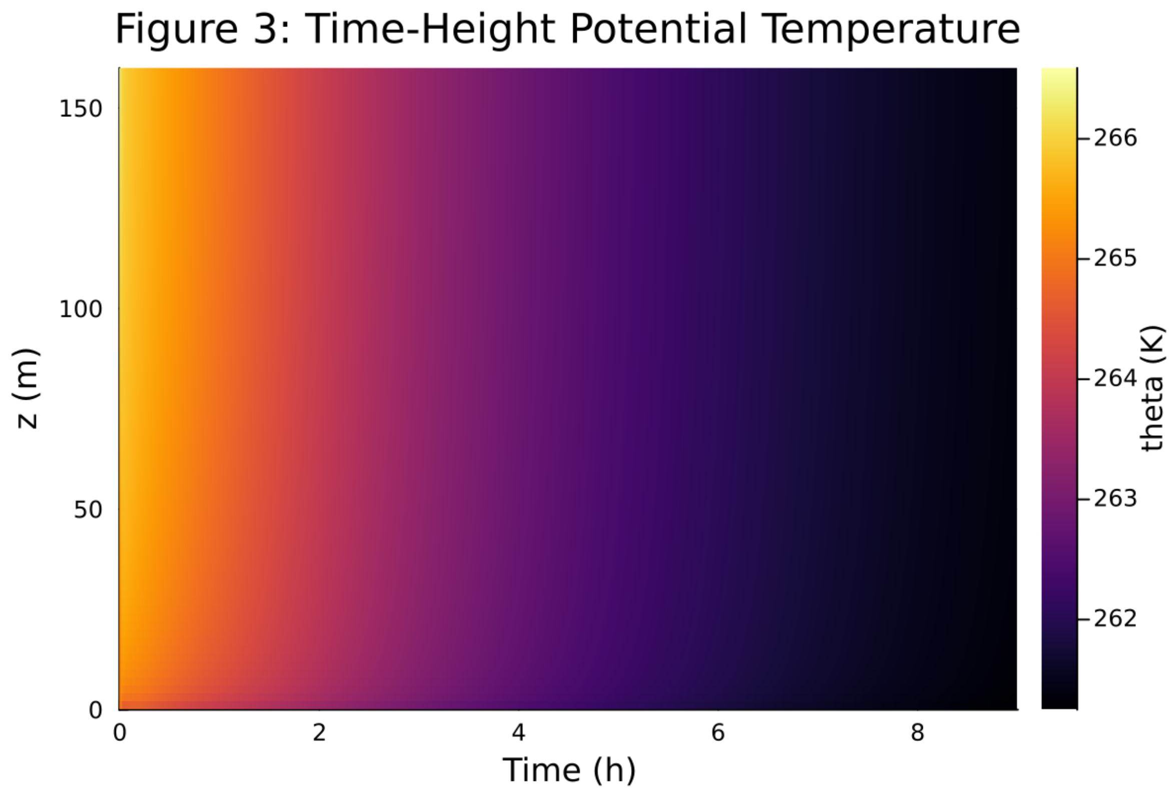

Figure 27: Time-height contour of potential temperature and inversion growth

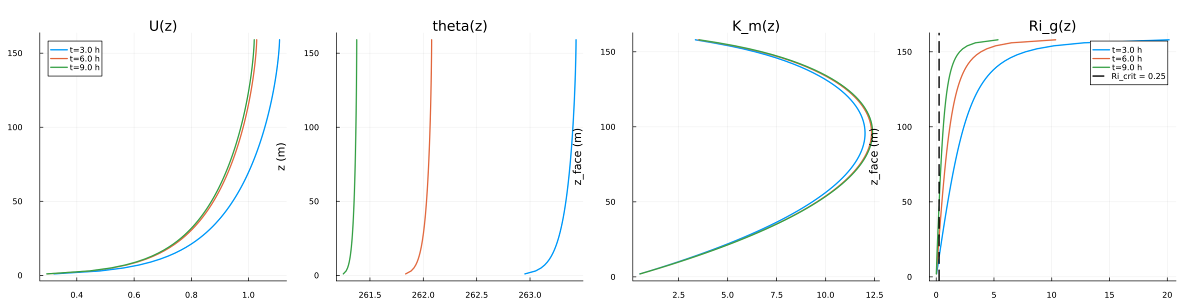

Figure 28: Vertical profiles of U, theta, and Km at representative times

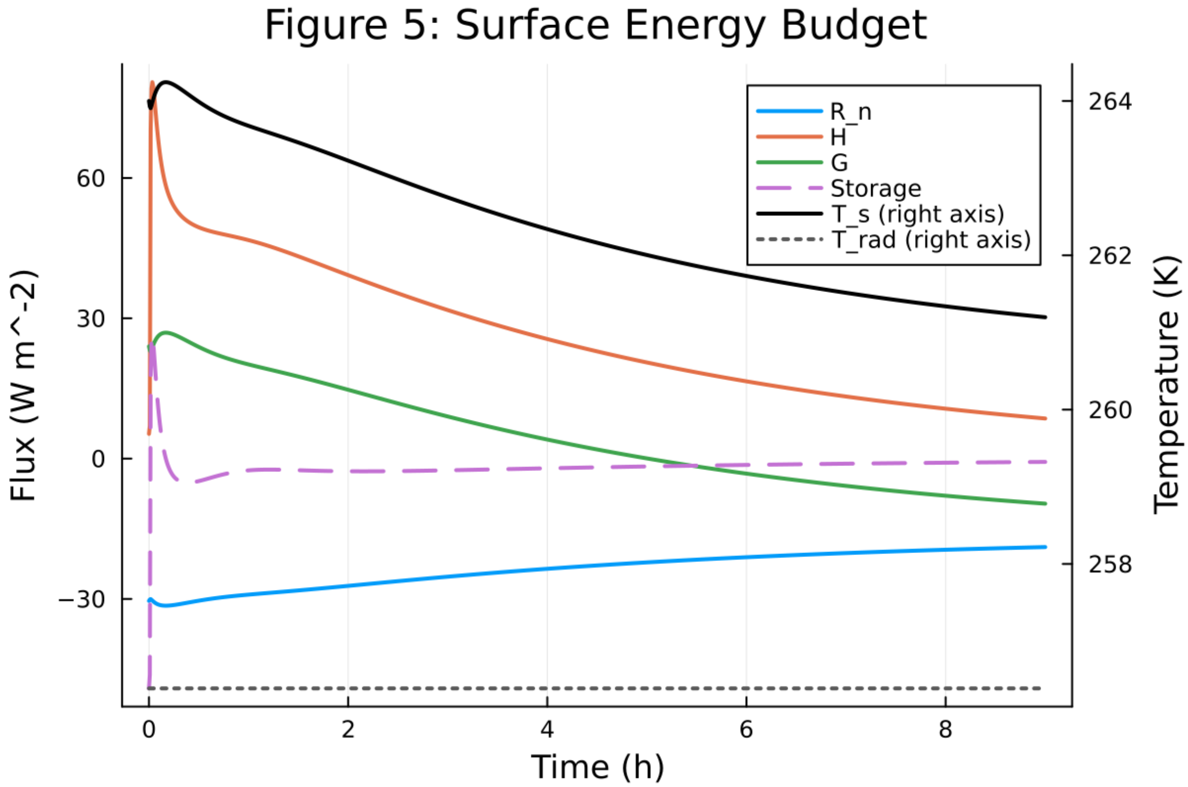

Figure 29: Surface energy budget components (R_n, H, G, storage)

Table 5: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	10.000000 $\mathrm{m\,s^{-1}}$
Meridional geostrophic wind	V_g	0.000000 $\mathrm{m\,s^{-1}}$
Surface roughness (momentum)	z_{0m}	n/a m
Surface roughness (heat)	z_{0h}	n/a m
Coriolis parameter	f	1.000e-04 $\mathrm{s^{-1}}$
Reference potential temperature	θ_a	280.000000 K
Deep soil temperature	T_{deep}	276.000000 K
Downward longwave forcing	$R_{\downarrow}$	300.000000 $\mathrm{W\,m^{-2}}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{\mathrm{min,surf}}$	0.001000 $\mathrm{m^2\,s^{-1}}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 $\mathrm{s^{-1}}$

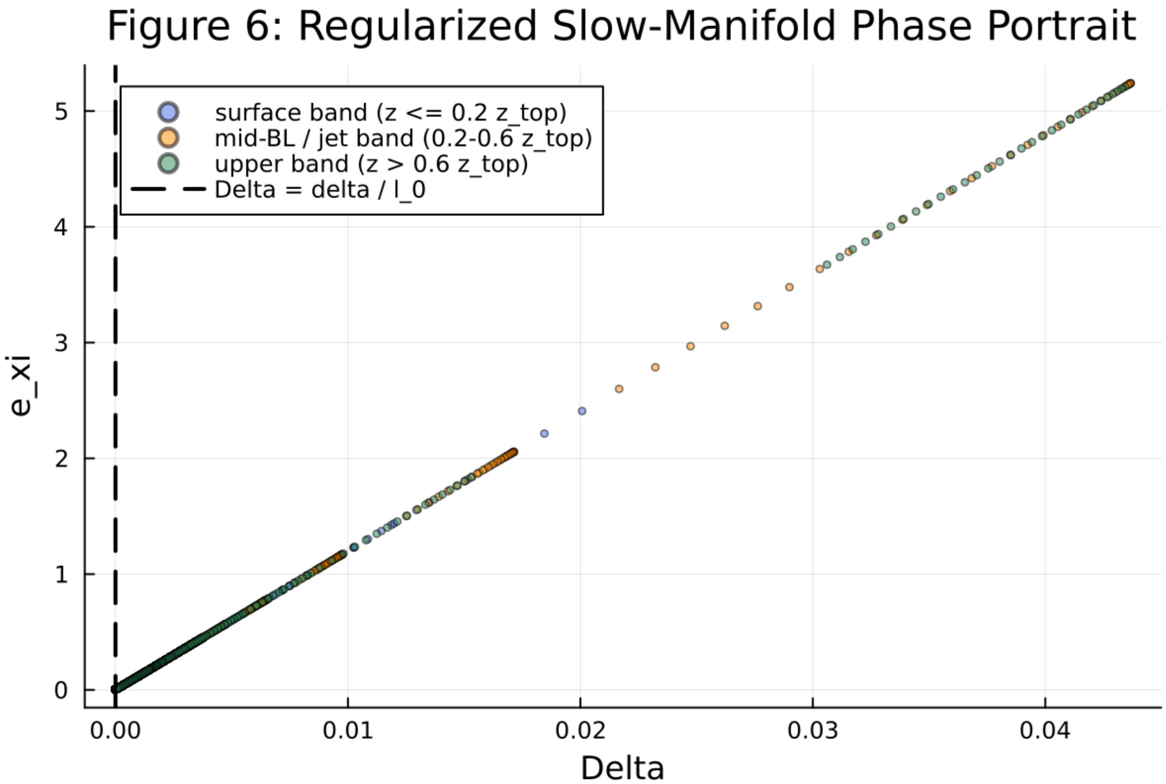

Figure 30: Phase portrait on regularized manifold (Delta vs e__xi)

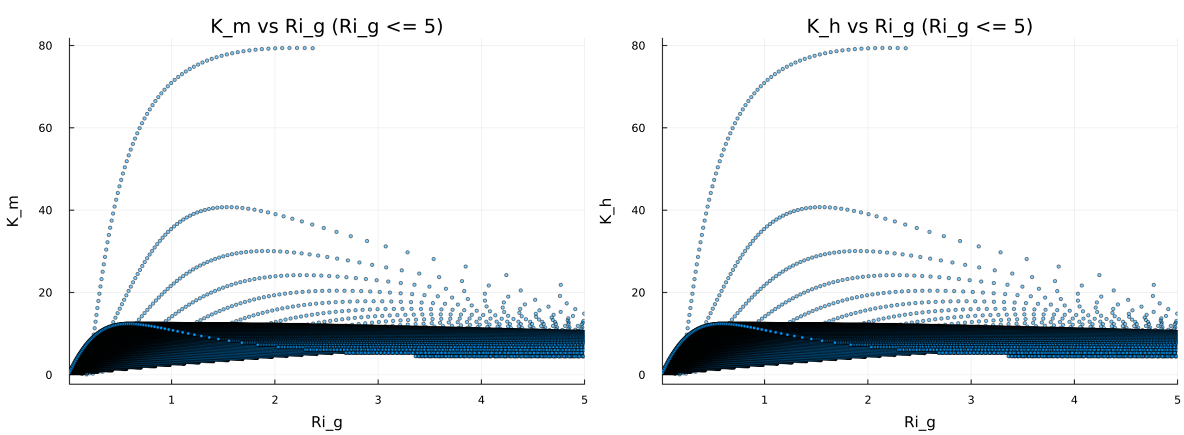

Figure 31: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

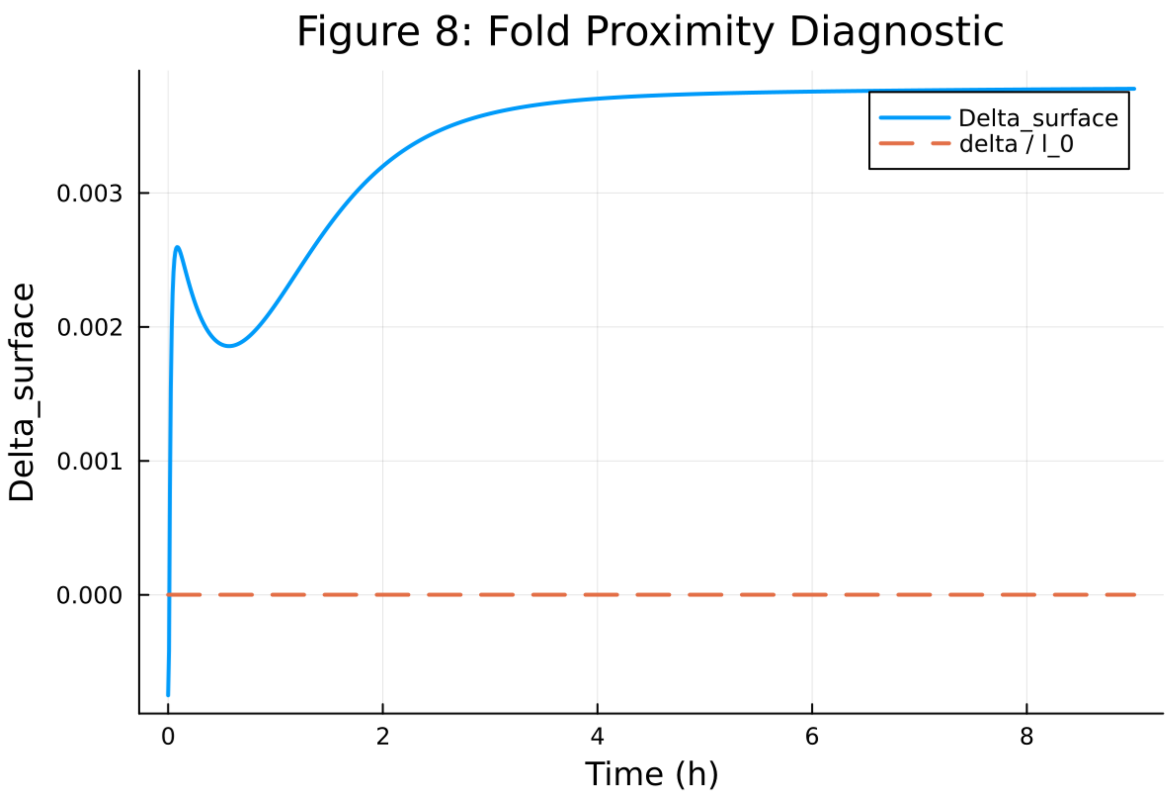

Figure 32: Fold proximity diagnostics versus time

11.2 Forcing and Boundary Conditions

11.3 Numerical Verification

- Maximum surface energy closure error: $0\,\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.184562, 108.426452]\ \mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [0.007283, 43.999865]$

11.4 Generated Artifacts

- Time-series CSV: `results/idealized_sbl/time_series.csv`
- Diagnostic summary JSON: `results/idealized_sbl/summary.json`
- Payload JLD2: `results/idealized_sbl/payload.jld2`

11.5 Figure Manifest

1. Time series of T_s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times

5. Surface energy budget components (R__n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e__xi)
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

11.6 Included Plots

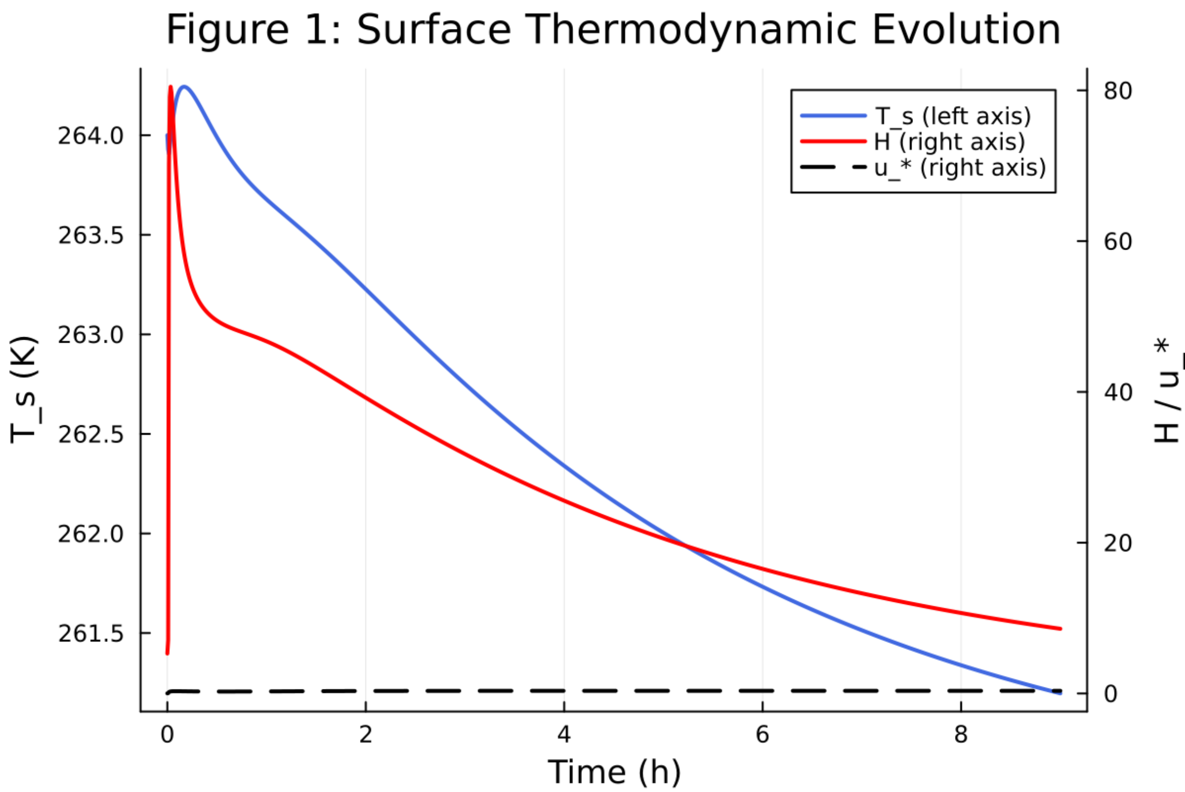

Figure 33: Time series of T__s, sensible heat flux, and friction velocity

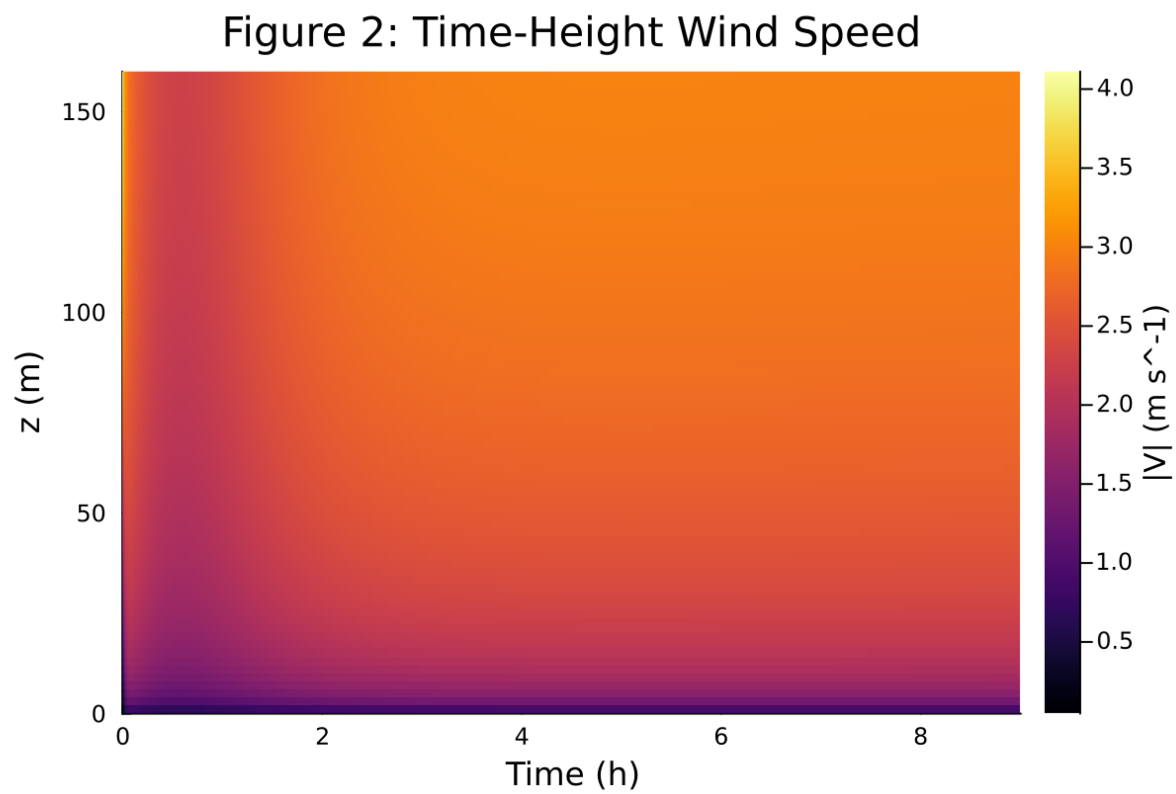

Figure 34: Time-height contour of wind speed with LLJ evolution

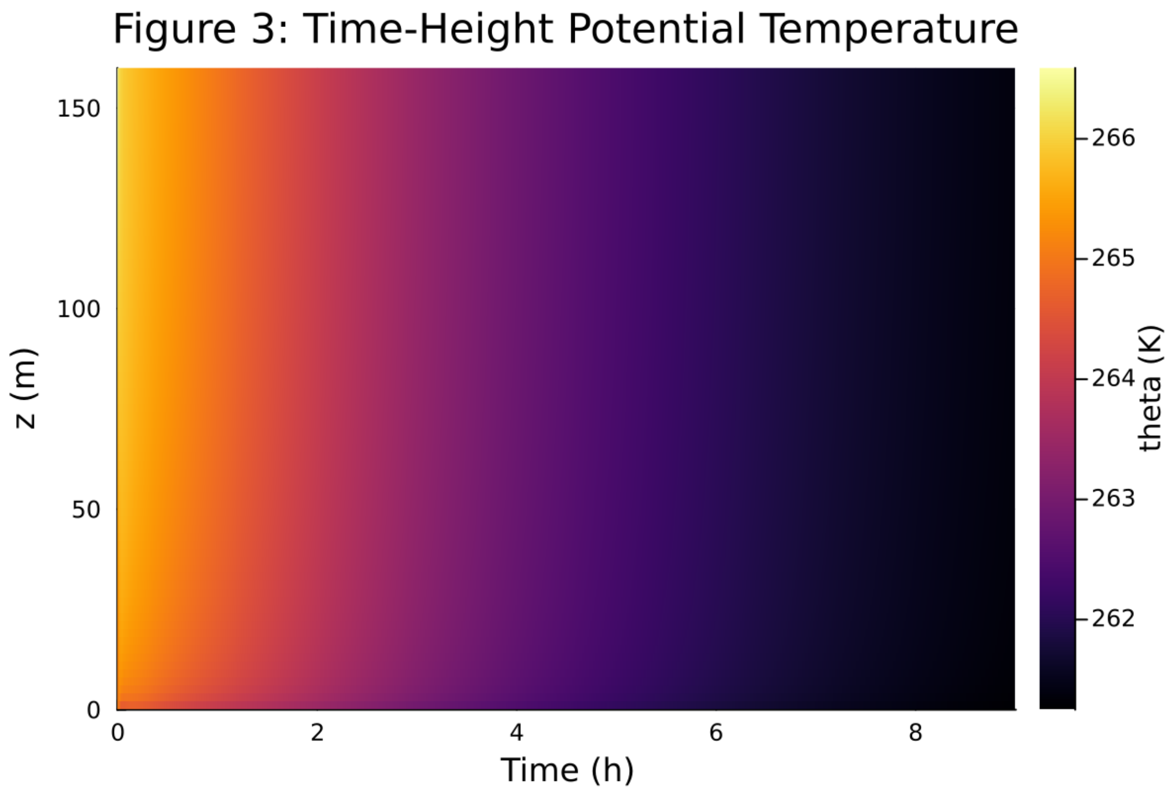

Figure 35: Time-height contour of potential temperature and inversion growth

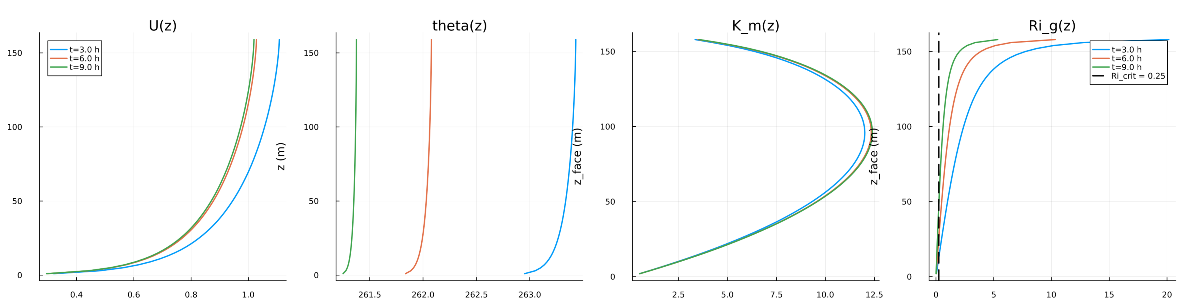

Figure 36: Vertical profiles of U, theta, and Km at representative times

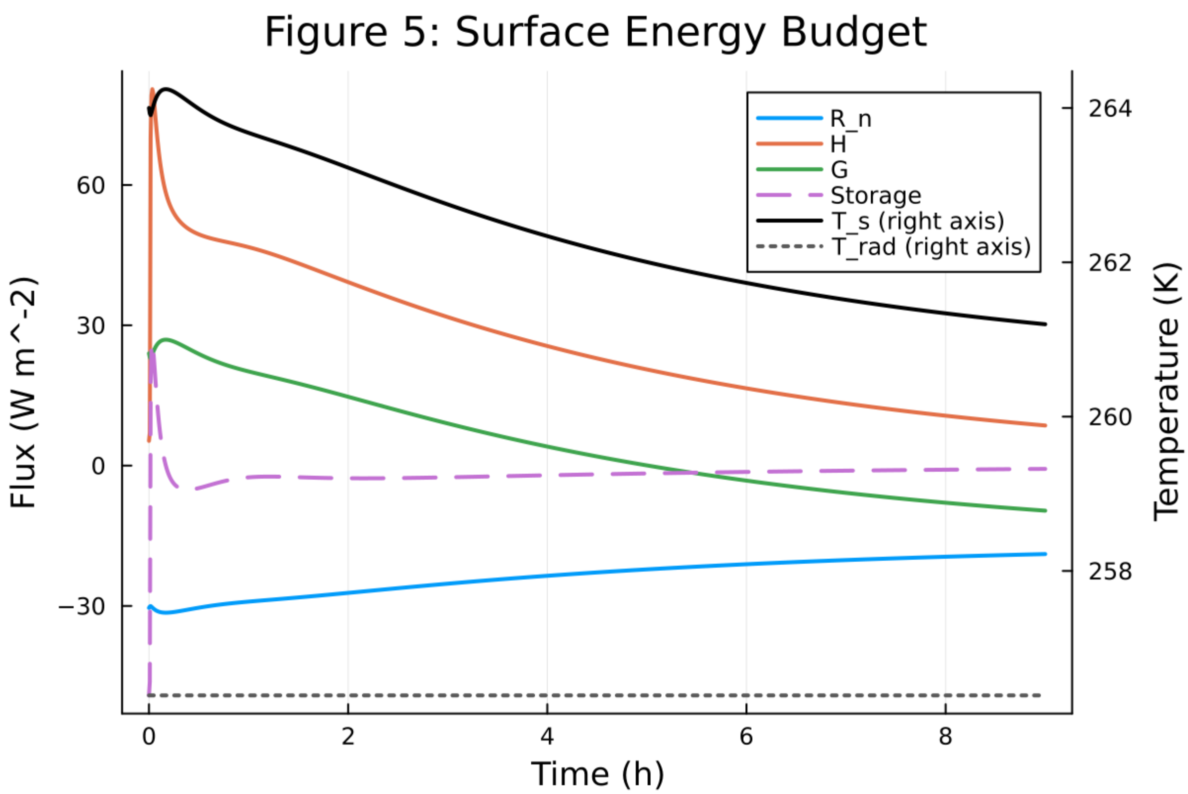

Figure 37: Surface energy budget components (R_n, H, G, storage)

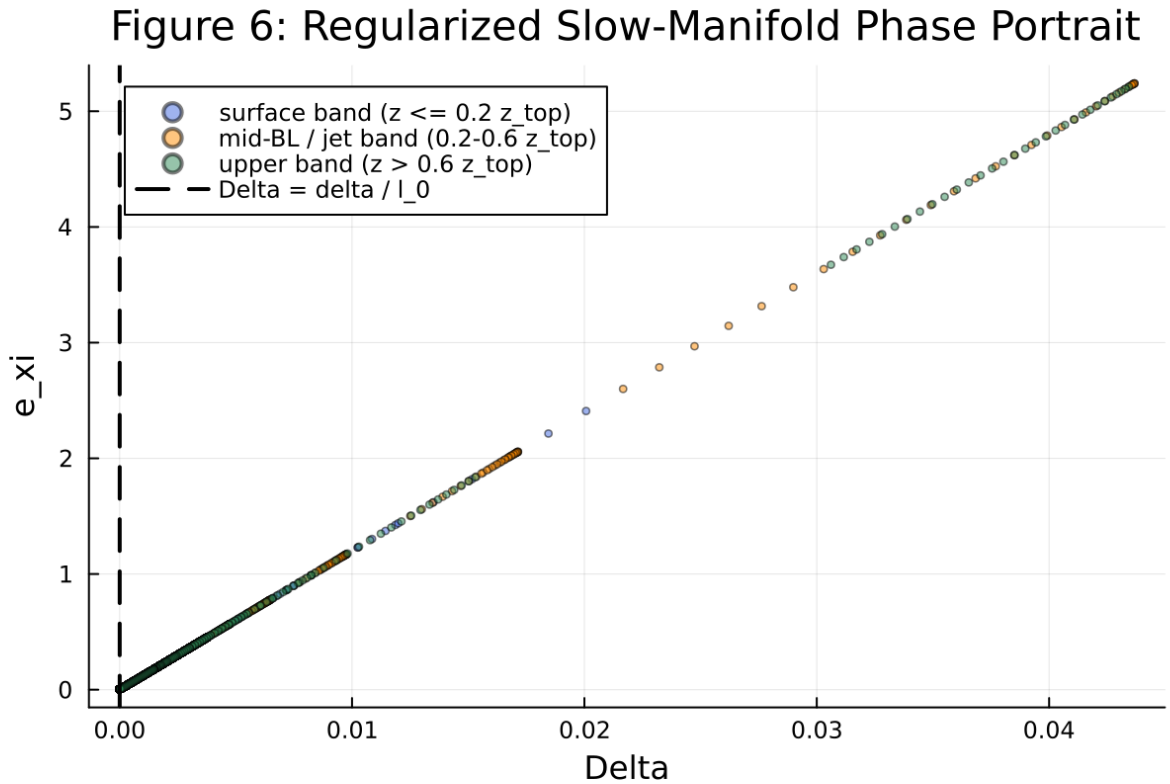

Figure 38: Phase portrait on regularized manifold (Delta vs e__xi)

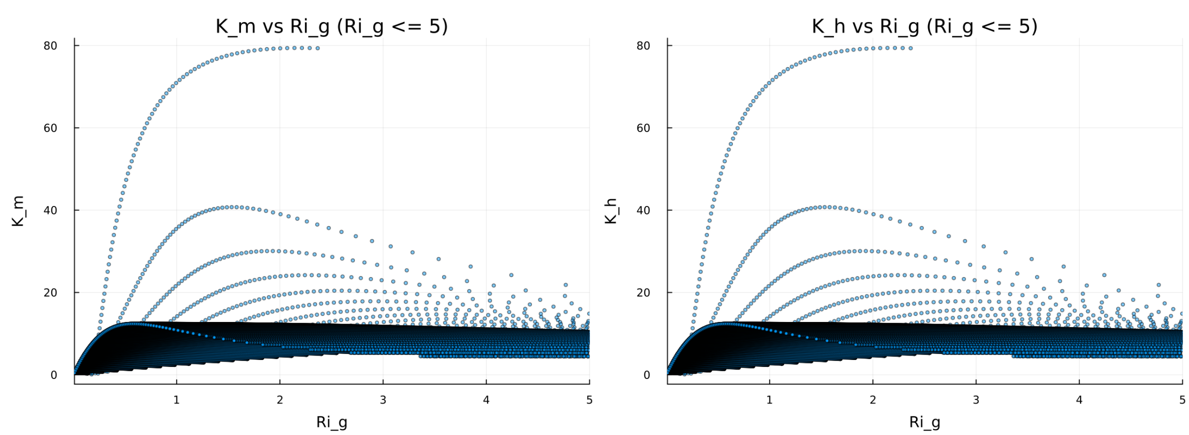

Figure 39: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

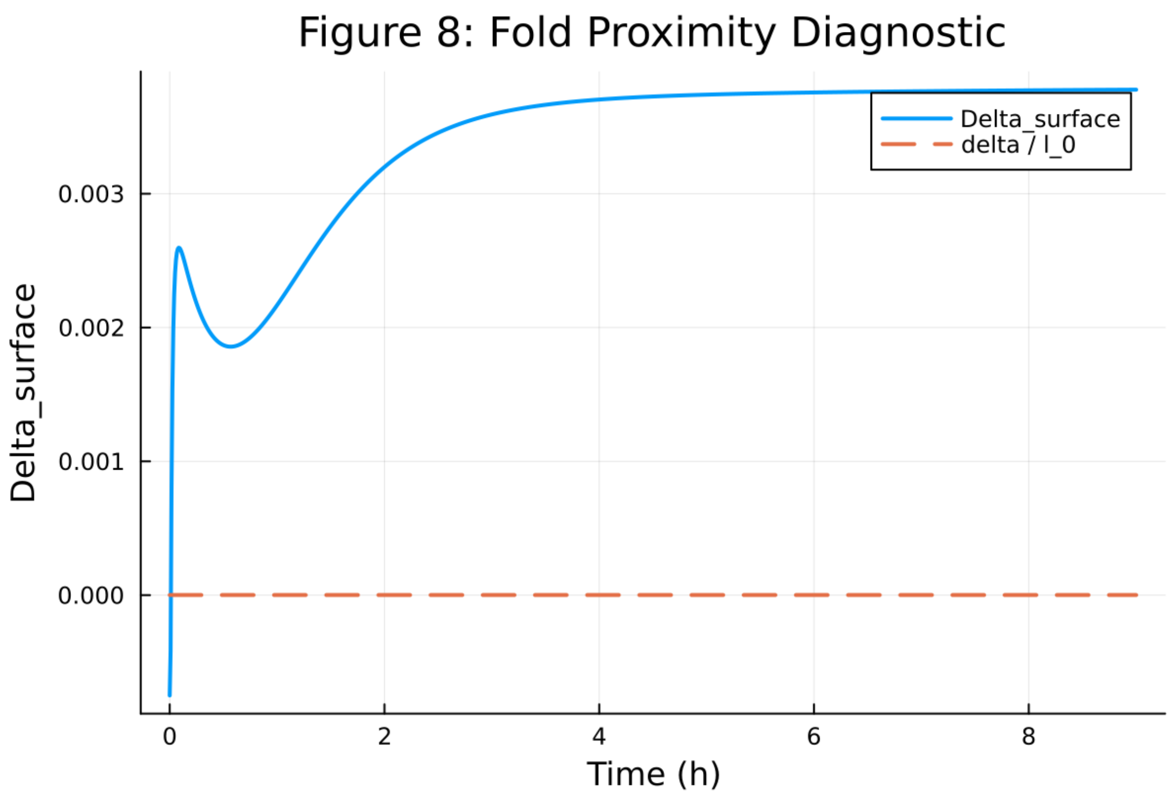

Figure 40: Fold proximity diagnostics versus time